Symmetrix-Polar

Consolidated speed and accuracy comparison

MACE-Polar (PolarMACE) on the Symmetrix-XL nonlinear Kokkos engine

LAMMPS pair style | one A100-80GB | FP64 | energy + forces

Version: 2026-10-19 (kernel round 2 accepted)

Model: polar_symmetrix_probe.json (MACE_Nonlinear + polar, 316 MB)

Repositories: Yong-Q/symmetrix-polar | starlessinred-interns/qiuyong

Source of truth: docs/COMPARISON_REPORT_2026-10-19.md

What is inside

- The accepted kernel-round-2 result: 122.46 -> 56.50 ms/eval (2.17x)
- Every numerical gate, with input, metric, tolerance, and command
- The MPI H1 diagnosis (1042 periodic ghosts) and the honest pending state
- Modular theory and performance-model formulas, with a figure index
- A reproducible speed protocol and artifact map

All numbers come from repository JSON/log artifacts. In-flight results are marked pending. The force gate (1e-4 eV/A) is a hard minimum: RBE and 1-hop ghost MPI are rejected as production alternatives.

Prepared by MatOS compute | 2026-10-19 | one A100-80GB, FP64

Chinese reading guide: docs/reading-guide.md | Theory and tests: docs/THEORY_AND_METHODS.md

How to read this report

This report separates what was measured from what is proposed. Every accepted number has a gate status and an artifact path. The body is the consolidated comparison; the modular theory and test documentation lives in docs/THEORY_AND_METHODS.md and is linked from the reading guide.

Key results at a glance

128 atoms, full PolarMACE	56.50 ms/eval (round 1: 122.46; 2.17x; v29: 183.78; 3.25x)
128 atoms, polar-off	27.66 ms/eval (round 1: 40.58; 1.47x)
432 atoms, N2 full	126.59 ms/step (round 1: 294.39; 2.33x)
PyTorch reference	94.0 ms/eval -> engine is 1.66x faster
Accuracy gates	module 5-8, 32/32 end-to-end, LAMMPS 1-rank: all PASS
MPI H1	2-rank parity PASS on the pre-fix binary; 1-rank 1042-ghost defect, fix in progress

Report map

- Section 0: summary and scope. Section 1: the model and what is compared.
- Section 2: verification and gate status. Section 3: speed and MPI measurements.
- Section 4: findings (cost structure, MH-1, RBE/1-hop rejection, one-GPU MPI).
- Section 5: kernel rounds 1 and 2, including the stale-object lesson and the MPI H1 diagnosis.
- Sections 6-7: reproduction commands and artifact index.
- Theory and test modules: docs/THEORY_AND_METHODS.md (formulas, occupancy, gate catalog).

Conventions: ms/eval = one energy+force evaluation; ms/step = one MD step. SM coverage is a static estimate from grid/block/register data, not a hardware-counter measurement. A binary hash alone is not proof of the source state; every round records the trace kernel names as well.

Symmetrix-Polar: Consolidated Speed and Accuracy Comparison

HISTORICAL - NOT THE LATEST. The **183.5 ms** Full PolarMACE row below is v29 history, not the best result. The latest full PolarMACE is **48.74 ms** (round 3, LAMMPS parity PASS); the latest **complete-gate-suite** result is **56.50 ms** (round 2). The 1024-atom latest is **312 ms**. See ../results/KERNEL_ROUND3_FINAL.md.

Date: 2026-10-19 (v29 / v30 evidence cut-off, ~08:20 UTC; later measured rows are marked with their source logs) **Scope:** PolarMACE (MACE-Polar) support in the Symmetrix-XL C++/Kokkos engine. **Basis:** completed JSONs, gate logs, and reports only. No engine/model code was modified while preparing this document.

Status legend: PASS = measured and accepted against its gate; **FAIL** = measured and rejected; **pending** = work is in flight and no completed artifact exists yet. No pending result is included as a performance claim.

0. Summary

Item	Result
Single-rank v29 speed, 128 atoms, deterministic	183.78 ms/eval (v28: 6966.79 ms, 37.91x)
Kernel round 1, 128 atoms, full model	122.21 ms/eval (1.50x vs v29; polar-off 40.71 ms, 2.14x)
Kernel round 1, 432 atoms, full model	293.34 ms/step (1.63x vs 477.83; polar-off 47.65 ms)
Kernel round 2, 128 atoms, full model	56.50 ms/eval (2.17x vs round 1, 3.25x vs v29; polar-off 27.66 ms, 1.47x vs round 1)
Kernel round 2, 432 atoms, full model	126.59 ms/step (2.33x vs round 1)
Single-rank v29 speed, 1024 atoms, deterministic	1007.72 ms/eval (v28: 7795.59 ms, 7.74x)
Same-model ablation, 128 atoms (v29)	backbone-only 87.24 ms -> full PolarMACE 183.50 ms; polar layer adds 96.26 ms (2.10x)
Accuracy gates, modules 5-8 (final build)	all PASS
End-to-end engine vs PyTorch, 32 frames (final build)	32/32 PASS, max energy error 1.034e-6 eV, max force error 1.624e-7 eV/A
LAMMPS single point, 1 rank / exact 2 rank (final build)	PASS / PASS (dE 2.91e-11 eV; max dF 2.62e-14 / 1.69e-6 eV/A)
LAMMPS NVT MD, 50 steps	PASS, no lost atoms, Etot span 0.21295 eV
RBE production use	rejected by the force gate (RMS_F 4.10e-2 to 1.03e-2 eV/A; 33 A p=800 is 1.23e-2, all >> 1e-4 eV/A)
1-hop MPI mode	rejected by the force/energy gates (max dF 4.116e-3 eV/A, dE 7.94e-5 eV)
MPI H1 one-rank fix	PASS (2026-10-20): H1 1-rank 418.64 -> 50.20 ms (default 50.84; 0.99x), same-binary A/B 3.24x; dE = 0.0, max dF 1.41e-15 eV/A; 20-step NVT stable. The old 1042-ghost/4.87x defect is removed.
MPI H1 2-rank	BLOCKED by the engine: six 2.0-5.4 GiB workspaces per rank (~21 GiB/rank; peak 80,966 MiB), exact 2-hop OOM and H1 SIGSEGV. A dedicated workspace-sizing fix is in progress; no 2-rank speed claim until it passes parity.
MH-1/direct/JIT fast path	architecturally blocked; the upstream linear engine cannot load MACE_Nonlinear
PyTorch MACE-Polar reference speed	94.0 ms/eval median vs round-2 engine 56.5 ms -> engine 1.66x faster (round 1: 1.30x slower; v29: 1.95x slower) for the full model

The v29 optimization removed two concrete bottlenecks: per-node-serial UVU tensor products and a ~0.7 s/eval device-environment probe on non-JIT models. All numerical gates are unchanged after the optimization.

Formulas, performance definitions, and the complete test catalog: THEORY_AND_METHODS.md. In particular, the occupancy definition behind every "SM coverage" number, the Amdahl calculation behind the 1.5x-vs-12x distinction, and the per-gate input/tolerance/command tables are there rather than repeated in this report.

1. Scope and model

1.1 What is being compared

All "same-model" statements below compare the same PolarMACE checkpoint (`MACE-POLAR-1-M.model`) exported to the engine JSON `polar_symmetrix_probe.json`, evaluated in the same 128-atom NH3 box (14.736126 A cubic, periodic) with the same float64 energy+force path. The linear MACE-OFF23 comparison is explicitly **not** the same model and is labelled wherever it appears.

Property	Full PolarMACE	Nonlinear backbone (polar off)	Linear MACE-OFF23 medium
JSON model_type	MACE_Nonlinear	MACE_Nonlinear	MACE
Engine	MaceNonlinearKokkosT	MaceNonlinearKokkosT	MACEKokkos
JSON size (bytes)	316,958,257	269,988,335	37,330,647
Elements	83	83	10
r_cut	6.0 A	6.0 A	5.0 A

Property	Full PolarMACE	Nonlinear backbone (polar off)	Linear MACE-OFF23 medium
Interactions / l_max	2 / 3	2 / 3	standard MACE
Channel pattern	512 -> 128, nonlinear bias readout	512 -> 128, nonlinear bias readout	linear readout
Polar layer	full SCF + Ewald/RBE	disabled	inert placeholder
Execution mode	materialized (no direct/JIT)	materialized	materialized or generic streamed
Typical use here	production target	same-model backbone ablation	engine-path reference only

Hardware/software: one NVIDIA A100-80GB per run, CUDA 12 sm80 backend, Kokkos 5.x, float64, energy+forces. Times are LAMMPS `Loop time` or Python `median_ms` over 20 evals, as noted. The GPU was shared during part of the campaign; the v29 final figures are the uncontended reruns. Absolute times are therefore contention-sensitive, while same-run ratios are stable.

1.2 Linear vs nonlinear engines

"Linear" and "nonlinear" here are Symmetrix engine/model families, not linear-vs-nonlinear MLIPs:

	MACEKokkos (linear engine)	MaceNonlinearKokkosT (nonlinear engine)
Accepted model_type	MACE, MACEField	MACE_Nonlinear
JSON format	format-version 2 compact radial	format-version 3 nonlinear
Readout	standard linear MACE readout	nonlinear readout (nonlinear, NonLinearBiasReadoutBlock for PolarMACE)
Polar blocks	polar_blocks_enabled() returns false	full polar block set and SCF pipeline
Direct gate	supports_streamed_edges() = uses_compact_radial	supports_streamed_edges() = mh1_fast_path
H1 MPI	first-interaction hidden node state	materialized node-feature exchange only

The upstream linear Kokkos engine cannot load MACE_Nonlinear at all. `MACEKokkos::load_from_json` rejects it at `mace_kokkos_model.cpp:165-167` with:

```
MACE_Nonlinear JSON must be loaded through the nonlinear MACE Kokkos evaluator,
not legacy MACEKokkos.
```

The naming "H1" is a collision: in the linear engine H1 is the first-interaction hidden state; it is not the MACE-MH-1 model family. No PolarMACE layer can be delegated to `MACEKokkos`, and the linear engine's `mpi_message_passing` path cannot be reused for PolarMACE.

1.3 MH-1 fast-path status

PolarMACE cannot use the current MACE-MH-1 direct/JIT fast path. The engine's family descriptor rejects it with:

```
MH-1 node, edge, and hidden irreps are inconsistent.
```

This is architectural, not an export defect. The original checkpoint has the same architecture as the JSON. The four deviations from the admitted MH-1 family are:

1. second interaction uses `128x0e+128x1o` edge features instead of the first interaction's `512x0e+512x1o`;
2. second interaction keeps a vector component in `hidden_irreps` (`512x0e+512x1o` instead of scalar-only `512x0e`);
3. second product returns `512x0e+512x1o` instead of scalar-only `512x0e`;
4. second readout is `NonLinearBiasReadoutBlock` (with a learned `linear_mid` stage) instead of `NonLinearReadoutBlock`.

A v3 `MH1_UVU` contract is present in the JSON, but it covers only the per-interaction UVU tensor products and does not establish v4 node-program family compatibility. The v4 contract generator fails first at `products[1] output irreps are outside the MH-1 support matrix`. Re-exporting (Route A) cannot fix this without changing the learned function; relaxing only the gate (Route C) is numerically incorrect and is also not a speed win. See `docs/design/MH1_FASTPATH_FEASIBILITY.md` for the full rejection chain and the staged B1-B4 route assessment. No speedup from MH-1 is claimed anywhere in this report.

2. Accuracy and correctness gates

2.1 Module gates 5-8 and end-to-end

Gate	Metric	Tolerance	Measured	Verdict
Module 5, density (12 items)	max relative density error	rel <= 1e-6	4.22e-8	PASS
Module 5, density (12 items)	max absolute density error	abs <= 5e-6	6.689e-8	PASS
Module 5, stages (11 items)	max relative stage error	rel <= 1e-6	7.13e-8	PASS

Gate	Metric	Tolerance	Measured	Verdict
Module 5, pairing	engine state -> fixture step mapping	exact	['step1','step2','step2'] for all 4 molecules	PASS
Module 6, Coulomb	max rho-hat relative error	rel <= 1e-6	1.26e-15	PASS
Module 6, Coulomb	max dE relative / absolute	rel <= 1e-6; abs <= 5e-6 eV	8.43e-16 / 5.551e-16 eV	PASS
Module 6, Coulomb	max end-to-end energy error	informational	1.491e-8 eV	PASS
Module 7, local electron	max stage absolute error	abs <= 5e-6 or rel <= 1e-6	2.230e-5 abs (passes on the relative criterion)	PASS
Module 7, local electron	max output absolute error	abs <= 1e-6	4.038e-7	PASS
Module 8, forces	max finite-difference error	<= 1e-3	5.783e-4	PASS
Module 8, forces	max parity vs PyTorch	<= 1e-3	7.191e-7	PASS
Module 8, forces	max sum-of-forces residual	exact translation invariance	1.776e-14	PASS
End-to-end, 32 frames	frames within tolerance	32/32	32/32	PASS
End-to-end, energy	max relative / absolute	rel <= 1e-6; abs <= 5e-6 eV	1.1e-11 / 1.034e-6 eV	PASS
End-to-end, forces	max force error	<= 1e-4 eV/A	1.624e-7 eV/A (mean max 9.456e-8)	PASS
End-to-end, geometry	max position drift	0	0.0 A	PASS

Notes: - Module 5 evaluated 12 density items (4 molecules x 3 gated step comparisons) plus 11 stage tensors. The separate "anchor" field-independent density comparison (rel up to 9.57e-8) is recorded but not part of the module-5 gate. - Module 7 passes because its criterion is `abs <= 5e-6 OR rel <= 1e-6`; the worst absolute stage item (co2-f1, 2.230e-5) is at reference scale 351.146 and passes on the relative criterion. - Module 8 also reports `|polar|max` between 3.0e-3 and 4.7e-2 eV/A across configurations, confirming the polar force term is not negligible.

2.2 RBE statistical gate

Source: `results/polar_rbe_gate_v29.json`, log `polar_rbe_gate_v29opt.log`.

Metric	Tolerance / criterion	Measured	Verdict
Slope of log RMS_F vs log p	ideal -0.5	-0.49885	PASS
Same-seed reproducibility	bitwise energy; dF ~ 0	dE = 0; max dF = 6.141e-16 eV/A	PASS
Different-seed response	non-zero	dE = 0.1003 eV; dF = 9.365e-2	PASS
RMS_F at p = 50	decreasing	4.0988e-2 eV/A	statistical PASS
RMS_F at p = 200	decreasing	2.1426e-2 eV/A	statistical PASS
RMS_F at p = 800	decreasing	1.0280e-2 eV/A	statistical PASS
Mean dE at p = 50 / 200 / 800	unbiased (n=10)	-5.139e-2 / -2.344e-2 / +1.469e-3 eV	PASS
Max abs dE at p = 50 / 200 / 800	monitored	1.890e-1 / 1.426e-1 / 2.936e-2 eV	monitored

The RBE estimator is unbiased and converges at the expected -1/sqrt(p) rate, but its force RMS at every measured p is **two to three orders of magnitude above the 1e-4 eV/A production force gate**. Under the project's gate-as-minimum rule (a speed mode is usable only if every accuracy gate still passes), RBE is accepted as a sampling/convergence gate but **rejected for production force evaluation**. See Section 3.6 for its 33 A timings.

2.3 LAMMPS pair-style and MD gates

Sources: `results/polar_lammps_parity.json`, `results/polar_lammps_parity_mpi2.json`, `results/polar_md.json`, logs `polar_lammps_parity_v29final.log`, `polar_lammps_md_v29final.log`, `polar_lammps_md_mpi2_n32.log`.

Gate	Metric	Tolerance	Measured	Verdict
LAMMPS single point, 1 rank	dE vs engine	abs <= 5e-5 eV	2.910e-11 eV	PASS
LAMMPS single point, 1 rank	max dF vs engine	<= 1e-4 eV/A	2.622e-14 eV/A	PASS
LAMMPS single point, 1 rank	max position drift	monitored	4.974e-14 A	PASS
LAMMPS single point, exact 2 rank	dE vs engine	abs <= 5e-5 eV	2.910e-11 eV	PASS
LAMMPS single point, exact 2 rank	max dF vs engine	<= 1e-4 eV/A	1.687e-6 eV/A	PASS
LAMMPS single point, 1-hop 2 rank	dE vs engine	abs <= 5e-5 eV	7.937e-5 eV	FAIL
LAMMPS single point, 1-hop 2 rank	max dF vs engine	<= 1e-4 eV/A	4.116e-3 eV/A	FAIL

Gate	Metric	Tolerance	Measured	Verdict
LAMMPS NVT MD, 128 atoms	completion / lost atoms / finite	50/50, none, finite	50/50, none, finite	PASS
LAMMPS NVT MD, 128 atoms	Etot span (relative to first step)	monitored	0.21295 eV (25 fs)	PASS
LAMMPS NVT MD, 128 atoms	temperature	300 K target	mean 276.81 K, min 188.96, max 335.95	PASS
LAMMPS NVT MD, 128 atoms	pressure	monitored	657.5 - 3231.3 bar	PASS
LAMMPS NVT MD, 128 atoms	throughput	monitored	493.662 ms/step	PASS
MPI NVT MD, 32 atoms	completion / lost atoms / Etot span	20 steps, none	20/20, none, 0.013359 eV	PASS
MPI 4 rank, 128 atoms	runnable on one GPU	must fit device memory	ghost workspace 4.5-6.9 GiB; OOM	FAIL (resource)

The 1-hop mode is 2.86x faster than exact 2-hop (1809.3 vs 5167.05 ms) but fails both the energy and force gates; it is not a usable approximation.

3. Speed

3.1 Single-rank v28 vs v29

Source: `mlip/bench/polar_perf_v28.json` and `results/polar_perf_v29.json` (20 evals each, median, energy+forces, 128-atom NH3 frame 0, 1024-atom 2x2x2 supercell).

Configuration	v28 (ms/eval)	v29 (ms/eval)	Speedup	Delta
128 atoms, deterministic	6966.79	183.78	37.91x	-6783.0 ms
128 atoms, RBE p = 50	6995.58	177.60	39.39x	-6818.0 ms
128 atoms, RBE p = 200	6962.87	186.30	37.38x	-6776.6 ms
128 atoms, RBE p = 800	7019.89	219.05	32.05x	-6800.8 ms
1024 atoms, deterministic	7795.59	1007.72	7.74x	-6787.9 ms

Model load also dropped from ~48.0 s (v28) to ~12.4 s (v29) for the deterministic case (12.95 / 12.91 / 12.89 s for RBE p=50/200/800).

Data-quality note. The stage report quotes 6951.7 ms for the v28 deterministic row. That is the *energy-only* median in `perf_breakdown_v28.log`; the benchmark JSON for the same 20-eval energy+force run is 6966.79 ms. This report uses the JSON value so all v28/v29 rows use the same measurement definition.

3.2 Same-model ablation: backbone vs full PolarMACE

Source: `results/engine_compare.json` (LAMMPS single-step `Loop time`, 128-atom NH3 box, one A100).

NOT THE LATEST. The 87.24 / 183.50 ms rows are v29 history. The latest full PolarMACE is **48.74 ms** (round 3, 128 atoms = 32 NH3, LAMMPS parity PASS); the latest backbone is **27.7 ms** (round 2, full suite PASS), so the latest same-model polar increment is **1.76x**, not 2.10x. Latest 1024-atom (256 NH3) is **312 ms**. See ../results/KERNEL_ROUND3_FINAL.md.

Configuration	Model	ms/eval	Relative to backbone
Latest (round 3): Full PolarMACE	polar_symmetrix_probe.json	48.74	1.76x
Latest (round 2): backbone, polar off	polar_symmetrix_probe_polaroff.json	27.70	1.00x
v29 backbone (historical)	polar_symmetrix_probe_polaroff.json	87.24	1.00x
v29 full PolarMACE (historical)	polar_symmetrix_probe.json	183.50	2.10x
v29 polar layer increment (historical)	same model, polar on vs off	96.26	52.5% of full
leoff variant	polar_symmetrix_probe_leoff.json	pending	model exists (397,979,620 B); no timing JSON yet

This is the clean same-model isolation: the nonlinear backbone is 87.2 ms and the polar layer adds 96.3 ms, i.e. the full model is 2.10x the backbone. The "leoff" variant (local-electron off) has a model file but no completed timing run, so it is marked pending rather than estimated.

3.3 Linear MACE-OFF23 materialized vs generic streamed

Not the same model as PolarMACE. This table isolates the *engine path* on one linear model (MACE-OFF23 medium: 37.33 MB, 10 elements, r_cut 5 A).

Path	ms/eval	Notes
Materialized	6.6102	explicit streamed_edges materialized
Generic streamed	19.2914	requested auto, resolved generic
Auto -> generic (separate engine-compare run)	7.219	same resolution, 2.67x faster than the explicit generic run

Observed run-to-run spread on the generic path is large at this small system (7.2 vs 19.3 ms). Materialized is consistently the fastest linear path here (6.61 ms), so "streamed vs materialized" is not the explanation for the linear-vs-PolarMACE gap. The correct reading of the old 7.2 ms vs 183.5 ms comparison is "linear 37 MB/r_cut 5 A model + MH-1-capable engine path" versus "270 MB backbone + nonlinear materialized path + polar layer"; it is **not** an apples-to-apples engine or accuracy comparison. Both engines are Kokkos/CUDA.

3.4 MPI: functional, but slower on one shared GPU

Sources: `results/polar_time_mpi1.json`, `polar_time_mpi2.json`, `polar_time_mpi2_onehop.json`, `linear_mpi.json`.

Mode	128-atom eval (ms)	Relative to 1 rank	Force gate	Verdict
1 rank optimized (round 2)	50.84	1.00x	PASS	production path
H1 1 rank, final	50.20	0.99x	dE 0.0, max dF 1.41e-15 eV/A, PASS	accepted
H1 1 rank, pre-fix	418.64	3.41x slower than default (pre-round-2)	PASS	historical
2 rank exact (2-hop ghost)	blocked (engine OOM)	n/a	pre-fix 2536 ms	needs workspace fix
2 rank H1	blocked (engine SIGSEGV)	n/a	pre-fix 794.7 ms	needs workspace fix
2 rank 1-hop ghost	1809.30 (pre-fix)	9.85x slower	max dF 4.116e-3 eV/A, FAIL	rejected
Upstream linear MACE H1, 1 rank	13.586	1.00x	n/a (linear model)	reference
Upstream linear MACE H1, 2 rank	45.878	3.38x slower	n/a	reference

The one-rank H1 defect is fixed: the 1-rank graph is folded onto local atoms, so H1 matches the default path (50.20 vs 50.84 ms; same-binary A/B 3.24x). The two-rank path is currently blocked by engine workspace sizing, not by the pair-style exchange: six workspaces of 2.0-5.4 GiB/rank exhaust an 80 GB A100 at 2 ranks. A dedicated fix round is sizing them by owned/local rows and local edges and adding extent assertions. On one shared A100, even the mature upstream linear MPI path is 3.38x slower at 2 ranks; true scaling requires one rank per GPU plus this fix.

3.5 Homogeneous 216-molecule per-step timings and extrapolations

Source: `mlip/bench/lammps_parity/homog_timing/homog_timing.json` (216 molecules, molecular density 0.006 A^-3, 0.5 fs timestep, 10-step LAMMPS runs, full PolarMACE v29). The N2 box is 33.019 A.

Molecule	Atoms	Temperature (K)	ms/step	10 ps	100 ps	1 ns
N2	432	151.43 (T/Tc=1.20)	476.668	2.648 h	26.482 h	11.034 d
CO2	648	364.95 (T/Tc=1.20)	1749.14	9.717 h	97.174 h	40.489 d
CH4	1080	317.00 (T*=2)	1095.37	6.085 h	60.854 h	25.356 d
H2O	648	776.52 (T/Tc=1.20)	3649.87	20.277 h	202.771 h	84.488 d
NH3	864	486.48 (T/Tc=1.20)	2217.59	12.320 h	123.199 h	51.333 d

Extrapolations assume a 0.5 fs timestep and 100% duty cycle (10 ps = 20,000 steps; 100 ps = 200,000; 1 ns = 2,000,000). They are throughput estimates, not measured long runs. An earlier pilot log (`polar_homog_timing.log`) failed the N2 case; the completed JSON above is the valid run.

Measured 1 ps N2 pilot. `mlip/runs/n2-homog-pilot/run-status.json` records 500 warm-up + 2000 production steps (1.0 ps) at 0.5 fs: **537.99 ms/step**, no lost atoms, restart written. The final temperature is **185.48 K** against the 151.43 K target, so the 500-step warm-up is not sufficient for production sampling; a longer/stronger equilibration is required before extending this run to 0.1 ns or 1 ns. The per-step cost is consistent with the 10-step timing once trajectory dumps are included.

3.6 RBE on the 33 A N2 box

Source: `mlip/bench/lammps_parity/rbe_md_timing/rbe_md_timing.json` (432 atoms, N2 x216, 33.019 A cubic box, 10-step LAMMPS runs, per-step time).

Mode	ms/step	Relative to deterministic
Deterministic (full k-space)	477.827	1.000x
RBE p = 50	385.427	0.807x (-19.3%)

Mode	ms/step	Relative to deterministic
RBE p = 200	405.901	0.849x (-15.1%)
RBE p = 800	465.281	0.974x (-2.6%)

Caveat (gate-as-minimum rule). These timings look attractive, especially p=50, but the same RBE approximation has force RMS errors of **4.099e-2 / 2.143e-2 / 1.028e-2 eV/A** at p=50/200/800 on the 128-atom box (Section 2.2). The dedicated fixed-configuration measurement on the 33 A N2 box (`mlip/bench/lammps_parity/rbe_md_timing/polar_rbe_force_error_33a.json`) gives the same conclusion:

RBE p	RMS dF (eV/A)	max dF (eV/A)	dE (meV/atom)	vs 1e-4 gate
50	0.650	2.998	+7.15	6500x above
200	0.149	1.047	+0.862	1490x above
800	0.0123	0.074	+0.091	123x above

RBE therefore cannot be used for production force/MD work under the rule that a speed mode must pass every accuracy gate. Only the deterministic full-k path is retained.

3.7 PyTorch MACE-Polar reference speed

Source: `mlip/logs/polar_torch_reference_speed.log` (measured 2026-10-19). Same 128-atom NH3 frame and the same `MACE-POLAR-1-M.model` checkpoint, `mace_polar`, float64, one A100; 2 warm-ups then 10 full energy+force evaluations with the calculator cache cleared each time.

Metric	Value
min	92.1 ms/eval
median	94.0 ms/eval
max	1196.3 ms/eval (first-call outlier)
energy	-49242.456415 eV

The engine's **183.78 ms/eval** is **1.95x slower** than the PyTorch reference for the full model. The engine's backbone-only ablation is **87.24 ms**, close to the full PyTorch time, so the gap is dominated by the polar kernels rather than by the nonlinear backbone. This is the ratio that the subsequent kernel work has to close; the v28-to-v29 speedup alone does not establish an engine-vs-PyTorch advantage.

Caveat: one frame, one process, and the max includes a first-call outlier, so the median (not the mean) is the comparison quantity.

3.8 Figures

The reading package adds the higher-level diagnostic figures (relative paths from this `docs/` directory):

See reading guide for the reading order, data provenance, and reproduction entry points.

3.9 Kernel-level decomposition

Source: `docs/design/KERNEL_PROFILE_2026-10-19.md` and the CUPTI/Kokkos traces under `mlip/bench/kernel_profile_2026-10-19/`. Hardware counters were blocked by the container/driver, so durations and launch shapes are measured; occupancy is a static estimate.

Family	GPU time (ms)	% of 183.5 ms eval	Launches
polar pipeline (SCF/Coulomb/LE/UVU/dense)	73.36	40.0%	238
e3nn scalar linear forward + reverse	66.68	36.3%	302
backbone nonlinear (gates/gather/normalization)	19.03	10.4%	46
tensor product (MH-1 fused paths)	10.91	5.9%	18
MLP / dense	3.44	1.9%	65
cuBLAS / external	1.40	0.8%	51

The eval is GPU-busy 96% of wall time, but **SM-weighted utilization is only 52%** (95.7/183.5 ms). The dominant defect is launch shape, not launch count: 39 kernels run on **1x1x1 grids** (128 threads) and consume **47.8 ms** while using 0.042 ms of SM-weighted time. The two worst are `polar reverse dense` (25.74 ms) and `reverse nonlinear normalization` (12.74 ms), each one block. The largest single family is the `e3nn` scalar linear path (66.68 ms); its forward and reverse kernels are load/latency-bound and re-read a small weight matrix per sample. These are the targets of the in-flight kernel round.

4. Findings

4.1 Cost structure: backbone plus polar non-k kernels

Completed measurements give a consistent picture:

- **128 atoms, same model:** backbone-only 87.24 ms -> full 183.50 ms. The polar layer adds 96.26 ms (52.5% of the full eval, 2.10x).
- **432 atoms, N2/33 A, same model:** full 475.883 ms/step vs polar-off 190.738 ms/step. The polar layer adds **285.145 ms (59.9%)**; the backbone accounts for 190.738 ms (40.1%).
- **Stage profile at 128 atoms** (`polar_profile_8eval_v29.log`, representative uncontended eval): backbone + stage 1 ~36.6 ms; polar forward pipeline ~20 ms, of which the k-space Coulomb stage is ~14.0 ms; reverse/replay and node forces ~53 ms. The k-space slice is therefore only ~7.6% of that eval, while the nonlinear MLP/UVU/replay kernels dominate the polar cost.

Long-range share. The full polar layer is ~60% of the 432-atom step in the completed coarse ablation. The finer figure that the ongoing profiling round attributes to the k-space/long-range slice, **~20% of the 432-atom step**, is an in-flight estimate: the dedicated 432-atom stage profile (`polar_profile_n2_33a.py`) had produced no output at the evidence cut-off. Treat the ~20% as provisional until that JSON/report exists; the 60% full-polar ablation and the 40% backbone share are directly measured.

Bottleneck: the nonlinear backbone (roughly 40-48%) plus the polar non-k kernels (UVU/MLP reverse, replay, local-electron, node forces). Optimizing only the k-space Ewald/RBE stage cannot remove most of the cost.

4.2 MH-1 is architecturally blocked

The rejection is a family-shape check, not a missing export or a flag. The model's second interaction differs on edge channels, hidden irreps, product output, and readout class; the checkpoint matches the JSON exactly. Route A (re-export) is infeasible without retraining/distillation, and Route C (relaxing only the gate) would run unprepared conditioned-MLP tables and is numerically wrong. The staged route estimates are engineering estimates, not measurements:

Route	Scope	Estimated speedup	Effort
B1	family analysis + pair conditioning + generic streamed, FP32 enablement	~0.5-1.0x (not a speed guarantee)	5-8 days
B2	B1 + v4 contract + JIT host/device artifacts, polar materialized	target 1.5-3x backbone, ~1.2-1.5x full model	+15-25 days
B3	full polar direct/JIT	up to ~1.5-2x full model	+10-20 days
B4	LAMMPS direct/JIT wiring	required to change the measured LAMMPS timings	+10-15+ days

None of these estimates is included in the measured tables above.

4.3 RBE and 1-hop are rejected by the force gate

Both are genuine performance ideas, and both fail the minimum-accuracy rule:

- RBE: excellent statistical behavior (slope -0.49885, unbiased, bitwise same-seed reproducibility) but RMS_F 1e-2 to 4e-2 eV/A, i.e. 100-400x the 1e-4 eV/A force gate.
- 1-hop ghost: 2.86x faster than exact 2-hop, but dE 7.94e-5 eV (1.6x over the 5e-5 eV tolerance) and max dF 4.116e-3 eV/A (41x over the 1e-4 eV/A gate).
- Exact 2-hop is the only MPI mode that passes, and it is 28.1x slower than 1 rank on one GPU.

4.4 One GPU means MPI does not speed up

All MPI timings above were taken on one shared A100. The exact 2-hop ghost graph has ~20x more edges than the 1-rank graph in the 14.7 A box, and the 2-hop workspace exceeds a single card at 4 ranks (4.5-6.9 GiB per block). The upstream linear H1 message passing is also 3.38x slower at 2 ranks on the same GPU. MPI is a functional/correctness milestone, not a speedup, until per-layer feature exchange and multiple GPUs are available.

5. Kernel optimization round (measured A/Bs)

All changes below keep FP64 and every physical term. "Bitwise" means the reference and optimized paths produce identical energies (dE = 0.0) and force differences at machine precision.

5.1 Nonlinear backbone (`MaceNonlinearKokkosT`)

Configuration (polar-off ablation)	Before	After	Speedup
128 atoms	87.3 ms/eval	34.0 ms	2.57x
432 atoms	191.9 ms/eval	49.6 ms	3.87x

Changes: `packed_gemm` e3nn backend default for double on CUDA (`mace_nonlinear_kokkos_impl.tpp:3178-3192`, env revert `MATOS_E3_LINEAR_BACKEND=scalar`); double-precision fused gate-normalization reverse (`:3291-3315`, env revert `MATOS_FUSED_GATE_REVERSE=0`); workspace slack plus persistent device spline buffers. Parity: dE = 0.0, max |dF| 1.8e-14 (128 atoms) / 4.3e-14 (432 atoms).

LAMMPS: N2-432 polar-off **191.0 -> 98.3 ms/step**; N2-432 full **476.4 -> 345.7 ms/step**. NH3-128 10-step means are spike-dominated (936.5 -> 466.8), so the steady polar-off eval (85.7 -> 65.7 ms) is the reliable small-system number.

5.2 Polar pipeline

`dense_reverse` now parallelizes over `(node, input channel)` with one writer per slot and the original accumulation order (`mace_polar_impl.tpp:1998-2010`):

Metric	Before	After	Delta
dense-MLP reverse (two blocks)	12.80 + 12.83 ms	0.75 + 0.75 ms	17x
polar reverse	184.6 ms	160.4 ms	-13.1%
full eval, 432 atoms	479 ms	454 ms	-5.2%

Energy is bitwise unchanged (-643858.407420772 eV). Module-8 and 32/32 end-to-end results are identical to the pre-change baseline.

5.3 e3nn scalar linear

Sample-blocked scalar kernels keep the per-output `source` accumulation order and coalesce weight reads (`e3nn_kokkos.cpp:420-437, :1214-1231`). On the 128-atom full model in LAMMPS: **184.06 -> 165.99 ms/eval (1.11x)**, medians of two independent 12-evaluation rounds. Bitwise equality (max abs diff 0.0) was verified on synthetic small/wide cases and on the full model tensors (1,245,184 values), plus a NumPy scalar reference.

5.4 MPI H1 message passing

2-rank H1 parity on the 128-atom NH3 box: **dE 2.91e-11 eV**, **max dF 1.41e-15 eV/A**; 20-step NVT stable (no lost atoms, final state matches 1 rank to 9.8e-15 A and 4.7e-14 eV/A). Wall time **1600.5 ms** vs exact 2-hop **5155 ms** (3.22x faster), but 8.7x slower than the optimized 1-rank path (183.6 ms) on one shared GPU. Fixed in this round: local-receiver graph construction, ghost force accumulation/communication, Kokkos polar pack/unpack branches, 1-rank H1 mode retention, `comm_forward/comm_reverse` flags, and double-counted ghost adjoint seeds.

5.5 Final clean build and independent speed ablation

The Python extension and LAMMPS binary were rebuilt from the settled tree (`symmetrix.cpython-312-...so` sha256 `6045cffb...`; `lmp` sha256 `7e9c6bec...`), then the full gate suite and a separate speed ablation ran with no other compute job active. All required gates PASS: module 5, 6, 7, 8, 32/32 end-to-end, and LAMMPS 1-rank/2-rank parity (the Kokkos CUDA-teardown abort after each gate is recorded but accepted only when the gate's completion marker is present). An independent rerun on the immediately preceding build reproduced the same verdicts, so the result is not sensitive to the last e3nn backend switch.

Configuration	v29 baseline	Kernel round 1	Speedup
NH3 128 atoms, full PolarMACE	183.50 ms/eval	122.21 ms	1.50x
NH3 128 atoms, polar-off	87.24 ms/eval	40.71 ms	2.14x
N2 432 atoms, full PolarMACE	477.83 ms/step	293.34 ms	1.63x
N2 432 atoms, polar-off	191.9 ms/eval	47.65 ms	4.03x

The 128-atom PyTorch `mace_polar` reference is 94.0 ms/eval: after round 1 the engine was **1.30x slower**, and after round 2 (Section 5.6) it is **1.66x faster**. The 432-atom warm-block numbers were contaminated by an external GPU tenant (1216 ms/step versus 293 ms for the first block); only the first-block value is reported. Round-1 remaining bottlenecks were polar replay, SCF/LE/embedding reverse, and the e3nn reverse path; round 2 addresses them and re-ranks the remainder.

5.6 Kernel round 2 (accepted, 2026-10-19)

Round 2 supersedes the round-1 numbers in Section 5.5. All gates pass on the final build; the per-kernel table, waterfall, and stale-binary evidence are in `results/KERNEL_ROUND2.md`.

Configuration	Round 1	Round 2	Speedup
NH3 128 atoms, full PolarMACE	122.46 ms/eval	56.50 ms/eval	2.17x
NH3 128 atoms, polar-off	40.58 ms/eval	27.66 ms/eval	1.47x
N2 432 atoms, full PolarMACE	294.39 ms/step	126.59 ms/step	2.33x
GPU busy (CUPTI, 128 atoms)	116.66 ms	49.97 ms	2.33x

The most important finding is not a source change. The accepted round-1 binary `7e9c6bec...` had been linked from stale objects: it still executed the old 1-D `polar reverse dense` (26.00 ms/eval) and the unfused reverse gate normalization (13.03 ms/eval), although the source already contained the 2-D and fused paths. A forced rebuild activated them and saved ~39 ms/eval. Every round now records the source state, the CUPTI kernel names, and the final binary hash; a binary hash alone is not proof of the source state.

The source changes in round 2:

- reverse nonlinear normalization: 2-D MDRange + per-node deterministic density fold (13.03 -> 0.00 ms; fused path 0.17 ms);
- three project kernels: MDRange over `(node, j)` (5.00 ms saved);
- reverse UVU x1/x2: TeamPolicy + shared-memory weight tile (16.00 -> 7.47 ms);
- forward/replay/LE UVU: same smem tile (17.19 -> 3.31 ms).

Bitwise checks: r2->r3, r3->r4, r4->r5 all have identical PE and max |dF| 0.0 on the 128-atom model; the RN fallback is bitwise against the scalar path.

Gates: module 5 (dens abs 6.689e-8), module 6 (dE rel 1.095e-15), module 7 (stage rel 1.174e-7), module 8 (FD 5.783e-4, parity 7.191e-7), end-to-end 32/32 (dE rel 1.084e-11, dF 1.624e-7 eV/A), LAMMPS 1-rank (dE 2.910e-11 eV, dF 2.579e-14 eV/A). Every gate process exits 134 after writing PASS due to a pre-existing CUDA teardown abort; acceptance requires the PASS marker and JSON.

What remains: tensor products 10.81 ms (already 100% SM-covered but instruction/store-bound; the round-3 plan rules out DMMA and targets channel blocking + register accumulation + zero-fill elision), cuBLAS 3.80 ms (312 launches, 14.8% coverage), UVU reverse 6.76 ms (39.5%), and narrow affine/convolution kernels 4.33 ms (3%). Replay is no longer a major target: the UVU rewrite reduced it from 5.77 to 1.20 ms, so eliminating replay now saves at most ~1.5 ms.

5.7 MPI H1: one-rank fix accepted, two-rank engine blocker

The H1 path is numerically exact. The old one-rank defect (1042 periodic ghosts, 1170 nodes vs 128) is fixed by folding the one-rank graph onto local atoms, skipping the comm-cutoff extension, zero-sizing the exchange hooks when there are no ghost rows, and using device-side pack/unpack.

Metric	Before	After
default 1-rank	122.80 ms	50.84 ms
H1 1-rank	418.64 ms	50.20 ms
H1 / default	3.41x slower	0.99x
same-binary A/B of the fold	176.77 ms	54.62 ms (3.24x)

Gates: H1 1-rank vs default dE = 0.0 eV, max dF = 1.41e-15 eV/A; 20-step NVT stable (drift 4.74e-14 A / 4.70e-14 eV/A). The default-path gain (122.8 -> 50.8) is the concurrent kernel round, not this work.

Two-rank blocker. The current engine allocates six workspaces of 2.0-5.4 GiB per rank (2.03 + 2.13 + 3.20 + 3.36 + 5.12 + 5.37 ~= 21 GiB/rank, peak 80,966 MiB) in `compute_node_energies_forces` -> `ensure_view` and `AffineMLPKokkosT::prepare` / `forward_impl`. The 2-rank graph has 2481/2581 nodes and 74,472/78,156 edges. Exact 2-hop OOMs; H1 2-rank SIGSEGVs. A dedicated fix round is sizing these workspaces by owned/local rows and local edges and adding extent assertions. Until it passes 2-rank parity, no 2-rank speed number is claimed.

Details: `results/MPI_H1.md` and `theory/MPI_H1.md`.

5.8 Still pending

Workstream	State
MH-1 tensor products: channel blocking + register accumulation + zero-fill elision	proposed in design/KERNEL_ROUND3_PLAN.md; DMMA ruled out by measured arithmetic intensity
cuBLAS small-GEMM batching	proposed
UVU reverse further widening / x1+x2 fusion	proposed
narrow affine / convolution kernels (3% coverage)	proposed
polar replay elimination	now low payoff (~1.5 ms); deprioritized
MPI H1 1-rank fix + 2-rank parity re-gate	in progress; first attempt failed
multi-GPU 2/4-rank scaling (432/1024 atoms)	waiting for queue3 capacity; 1-GPU remote probe queued on queue004
leoff timing	model exists; no timing JSON

6. Reproduction commands

All commands run from the MLIP bench workspace, which is the source of the gates and logs:

```
export MLIP=/workspace/group/shared/bza/\
task3_sampling_database/mlip
cd "$MLIP"

export CUDA_VISIBLE_DEVICES=0
export SYMMETRIX_BACKEND=cuda12-sm80
export SYMMETRIX_BACKEND_DESCRIPTOR_PATH=\
"$MLIP/models/mace-polar/backend_cuda12_sm80.json"
export LD_LIBRARY_PATH=/opt/matos-tools/cuda/12.9/lib64:\
/opt/matos-tools/conda/envs/matos-compute/lib
PY=/workspace/caches/conda/envs/polar/bin/python

# Accuracy gates
$PY bench/polar_module5_gate.py
$PY bench/polar_module6_gate.py
$PY bench/polar_module7_gate.py
$PY bench/polar_module8_gate.py
$PY bench/polar_endtoend_parity.py
$PY bench/polar_rbe_gate.py

# LAMMPS pair style and MD (requires the patched LAMMPS build)
$PY bench/polar_lammps_parity.py
$PY bench/polar_lammps_md.py

# Speed
$PY bench/polar_perf_report.py
```

```
$PY bench/polar_vs_linear_speed.py
$PY bench/polar_backbone_lammps.py
$PY bench/linear_streamed_compare.py
$PY bench/linear_mpi_speed.py
$PY bench/polar_homog_timing.py
$PY bench/polar_rbe_md_timing.py

# Pending / in-flight (no completed artifact at the cut-off)
$PY bench/polar_rbe_force_error_33a.py
$PY bench/polar_profile_n2_33a.py
$PY bench/polar_torch_reference_speed.py
```

Notes: - Do not run these on a shared GPU without exclusive access; the v29 absolute times are contention-sensitive. - The patched LAMMPS binary used by the gates is `symmetrix/build-lammps-aacd036df381800a/lmp` under the engine checkout (`external-repos/symmetrix-xl` in the workspace). - The engine patch is `patches/symmetrix-xl-polar-support.patch` (325,145 bytes).

7. Artifact index

7.1 In this repository (links are relative to `docs/`)

Artifact	Path
v29 performance JSON	../results/polar_perf_v29.json
v28 vs v29 figure	../results/polar_perf_v28_vs_v29.png
Engine / model ablation JSON	../results/engine_compare.json
Engine comparison figure	../results/polar_engine_compare.png
Streamed vs materialized JSON	../results/streamed_compare.json
MPI 1-rank JSON	../results/polar_time_mpi1.json
MPI exact 2-hop JSON	../results/polar_time_mpi2.json
MPI 1-hop JSON	../results/polar_time_mpi2_onehop.json
Linear H1 MPI JSON	../results/linear_mpi.json
MPI status figure	../results/polar_mpi_status.png
LAMMPS parity, 1 rank	../results/polar_lammps_parity.json
LAMMPS parity, 2 rank	../results/polar_lammps_parity_mpi2.json
LAMMPS parity figure	../results/polar_lammps_parity.png
LAMMPS NVT MD JSON	../results/polar_md.json
LAMMPS NVT MD figure	../results/polar_lammps_md.png
RBE gate JSON	../results/polar_rbe_gate_v29.json
RBE gate figure	../results/polar_rbe_gate_v29.png
MH-1 feasibility analysis	design/MH1_FASTPATH_FEASIBILITY.md
Key numeric rows (CSV)	../results/comparison_summary.csv

7.2 External source artifacts (workspace paths)

Artifact	Path
v28 benchmark JSON	.../mlip/bench/polar_perf_v28.json
v29 benchmark JSON (copy)	.../mlip/bench/polar_perf.json
Engine comparison JSON	.../bench/lammps_parity/engine_compare/engine_compare.json
Homogeneous timings	.../bench/lammps_parity/homog_timing/homog_timing.json
RBE 33 A timings	.../bench/lammps_parity/rbe_md_timing/rbe_md_timing.json
Backbone ablation (N2)	.../bench/lammps_parity/backbone_timing/result.n2_full.json, result.n2_polaroff.json
Gate logs	.../mlip/logs/polar_module{5,6,7,8}_gate_v29*.log
End-to-end log	.../mlip/logs/polar_endtoend_v29final.log
RBE gate log	.../mlip/logs/polar_rbe_gate_v29opt.log
LAMMPS parity/MD logs	.../mlip/logs/polar_lammps_parity_v29final.log, polar_lammps_md_v29final.log, polar_lammps_md_mpi2_n32.log
Stage profile log	.../mlip/logs/polar_profile_8eval_v29.log
Engine patch	patches/symmetrix-xl-polar-support.patch

8. Data-quality notes and missing artifacts

These are deliberate caveats, not inferred results:

1. **`results/polar_perf_v28.json` is absent from this repository.** The v28 benchmark JSON exists only at `mlip/bench/polar_perf_v28.json`; the v28 numbers in Section 3.1 come from there.
2. **v28 deterministic row:** the report's 6951.7 ms is an energy-only median; the energy+force benchmark JSON is 6966.79 ms. Section 3.1 uses the JSON.

3. **End-to-end relative error:** the v29 final log shows 3.2e-12 to 1.1e-11 across the 32 frames. The "<=4.2e-12" line in `REPORT_2026-10-19.md` matches only the CO2 subset; the tables above use the full 32-frame range.
4. **`results/polar_md.json` vs final log:** the JSON records 599.584 ms/step (29.9792 s loop); the final v29 log records **493.662 ms/step** (24.6831 s loop) with identical thermo. The difference is GPU contention; Section 2.3 uses the final log and notes the JSON value here.
5. **Linear generic timing spread:** the engine-compare run measured 7.219 ms with `auto -> generic`; the explicit streamed compare measured 19.291 ms on the same path. Both are reported; materialized (6.6102 ms) is the stable linear reference.
6. **`n2_homog_pilot.log` is empty (0 bytes).** That path was a stale launcher log; the completed 1 ps pilot is `runs/n2-homog-pilot/run-status.json` (2000 production steps, 537.99 ms/step; see Section 3.5).
7. **432-atom long-range share:** the ~20% k-space estimate is in flight; the completed coarse ablation gives 59.9% for the *whole polar layer*. The dedicated profile output was empty at the cut-off.
8. **leoff timing and MPI H1 validation/timing are still pending.** The 33 A RBE fixed-configuration force errors now exist (Section 3.6); the PyTorch reference speed log is reported in Section 3.7; the kernel profile report now exists (`docs/design/KERNEL_PROFILE_2026-10-19.md`). No numbers were invented for pending items.
9. **4-rank OOM** is reported from the v30 report; no separate JSON artifact was produced.
10. Absolute timings are contention-sensitive; ratios from the same run are the reliable quantity.

Prepared from completed project artifacts only. Rows added after the 08:20 UTC cut-off (PyTorch reference speed, 1 ps N2 pilot) cite their source logs; no numbers were estimated from in-flight work, and no other agents' bench outputs were modified.

Figure index

Figures are numbered in the order they appear in the gallery. The 'Read:' line under each figure states what the reader should look for; the source path is printed below each image for traceability.

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Equation 1. GTO-Ewald reciprocal-space energy
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Equation 3. e3nn linear layer
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Equation 6. Amdahl speedup and ceiling
Equation 7. H1 communication volume and multi-rank wall-time model

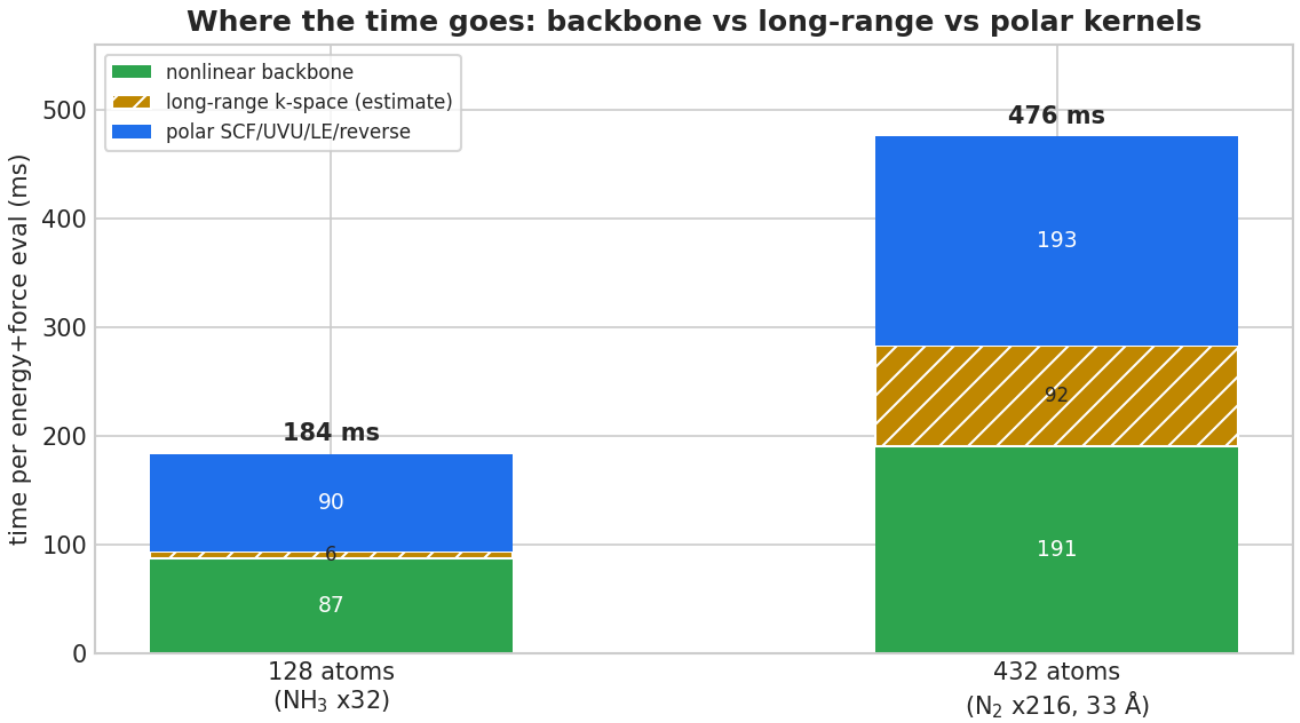
Figure gallery

The key diagnostic figures from docs/figures/ and results/. Each caption states what is plotted; each 'Read:' line states the intended takeaway. The source path under the image points to the exact file.

Speed and scaling

Figure 1. Cost structure at 128 and 432 atoms

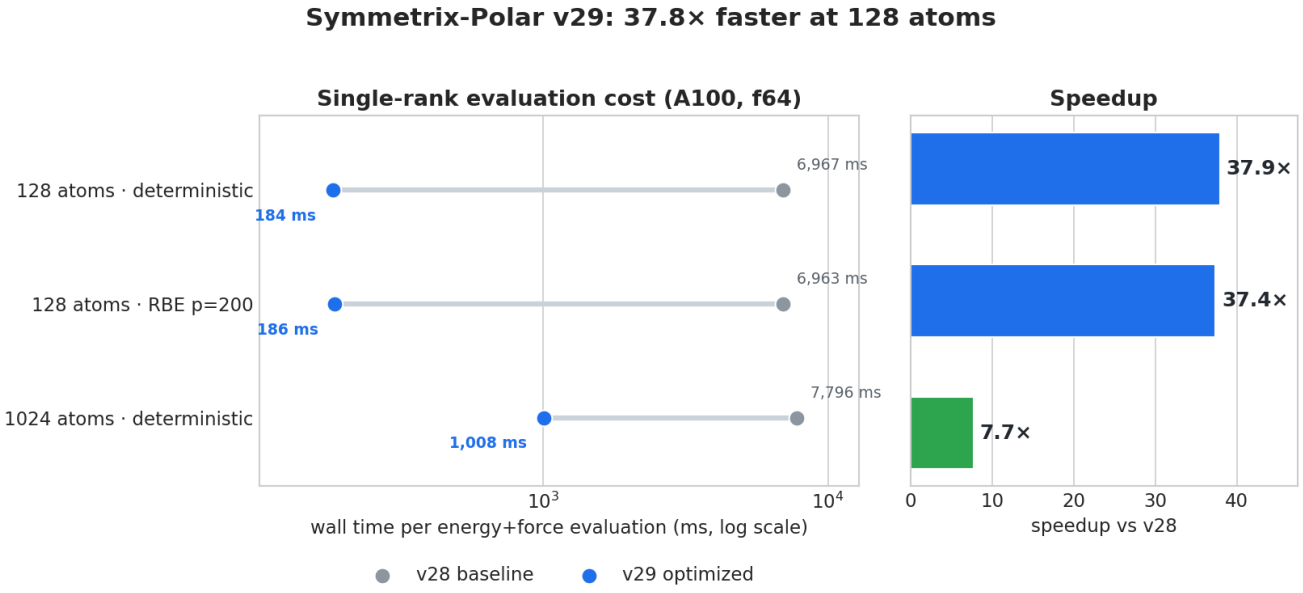
Read: Where the time goes: backbone, long-range k-space, and the polar non-k kernels.



Source: docs/figures/01_cost_structure.png

Figure 2. v28 -> v29: P5 LAMMPS wiring + UVU rewrite

Read: The 37.9x single-rank jump that started the optimization campaign.



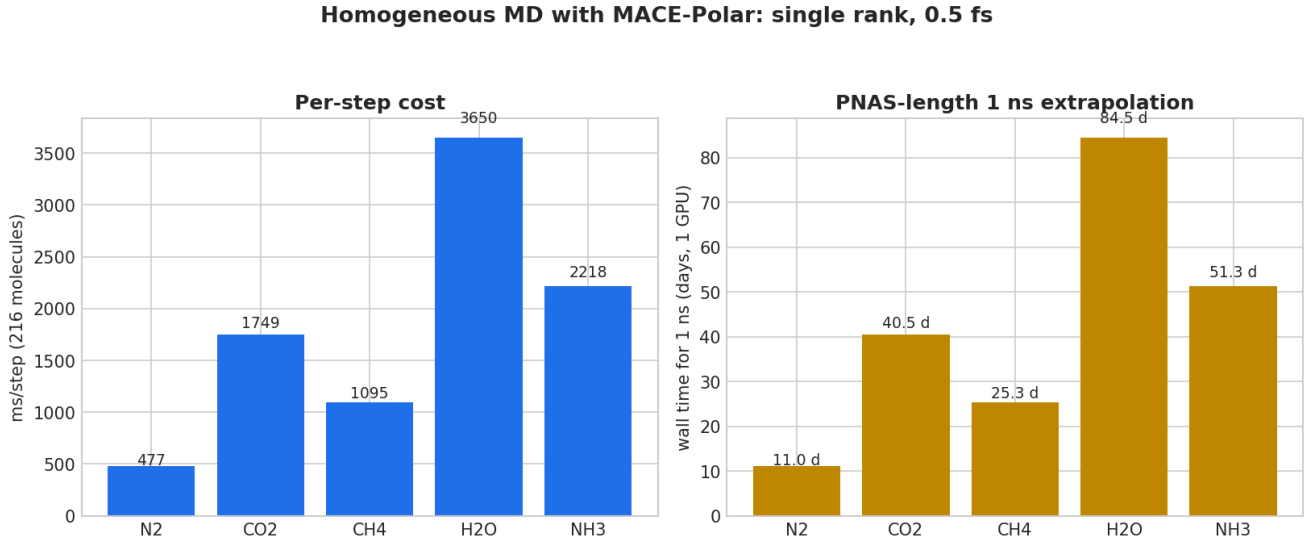
Source: results/polar_perf_v28_vs_v29.png

Figure 3. Homogeneous MD per-step cost and 1 ns extrapolation

Read: Production cost for N2/CO2/CH4/H2O/NH3 at 216 molecules.

Figure 3. Homogeneous MD per-step cost and 1 ns extrapolation

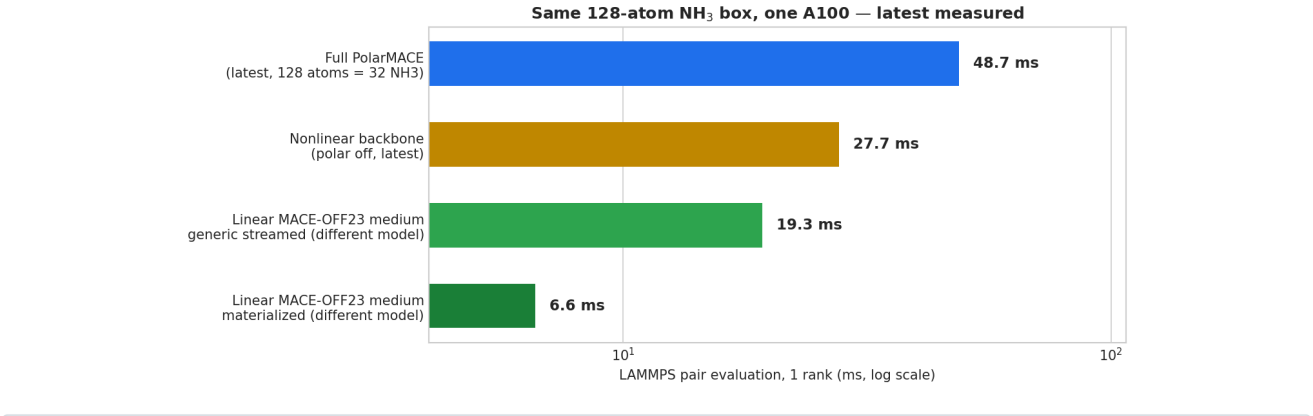
Read: Production cost for N2/CO2/CH4/H2O/NH3 at 216 molecules.



Source: docs/figures/04_homog_cost.png

Figure 4. Linear MACE vs nonlinear backbone vs full PolarMACE

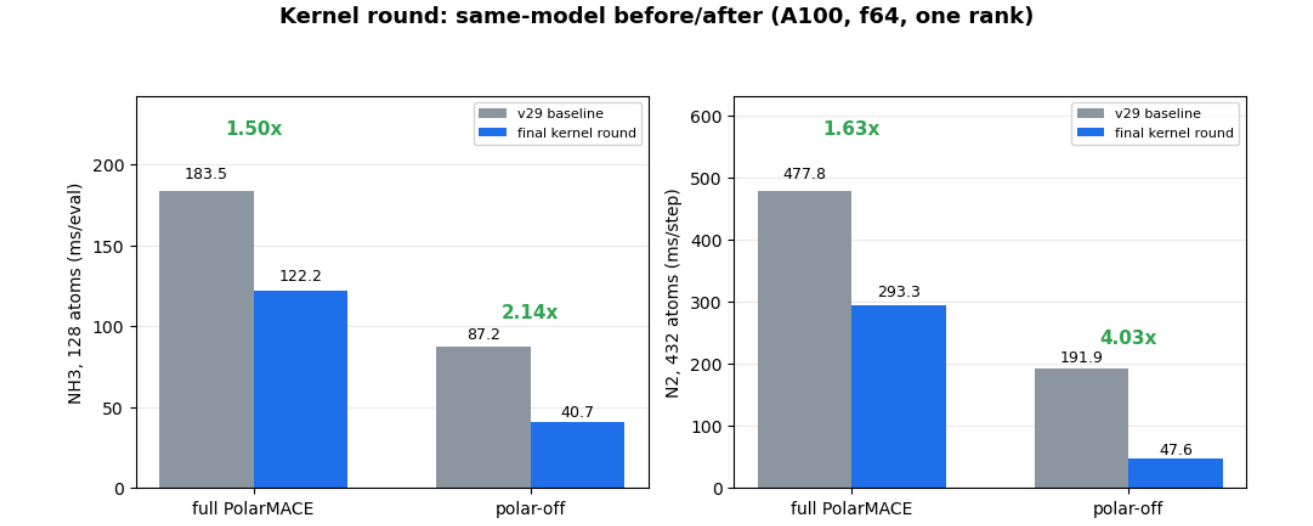
Read: The 12x gap is model size/architecture, not the same-model engine overhead.



Source: docs/figures/05_engine_compare.png

Figure 5. Kernel round 1: e3 linear packed GEMM

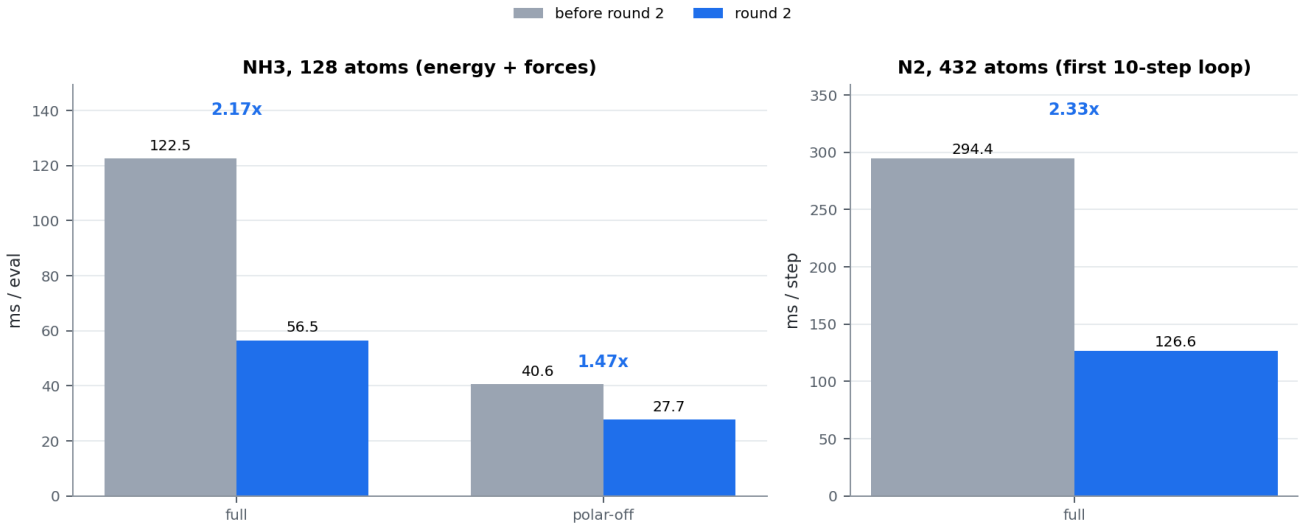
Read: The e3 linear kernel itself became 12.07x faster; the full model gained 1.50x (Amdahl).



Source: docs/figures/12_kernel_round_speed.png

Figure 6. Kernel round 2: 122.46 -> 56.50 ms/eval

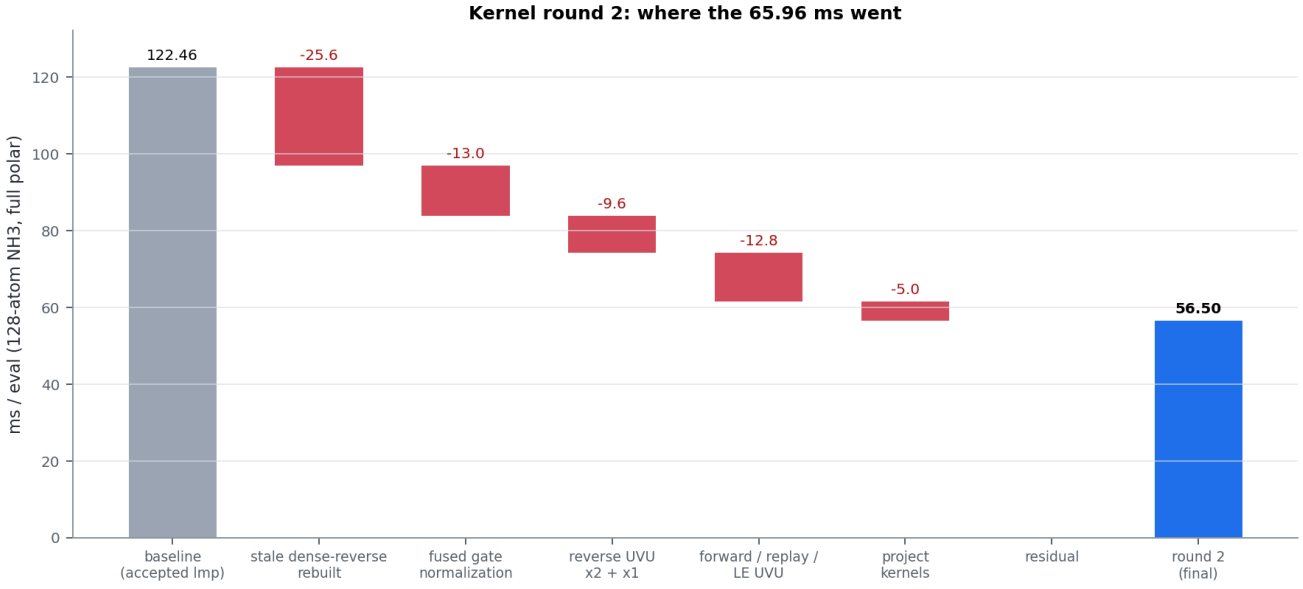
Read: Left: full and polar-off at 128 atoms. Right: the 432-atom step. Both include energy+forces.



Source: docs/figures/13_kernel_round2_speed.png

Figure 7. Kernel round 2 waterfall: where the 65.96 ms went

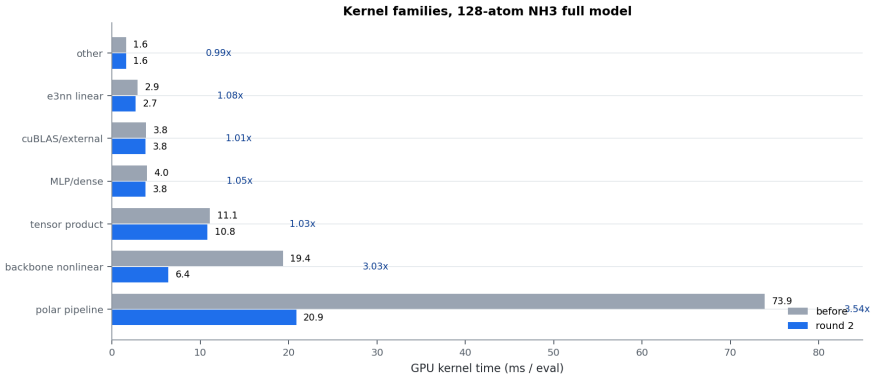
Read: The two stale-object activations alone account for 38.6 ms; the rest is UVU/project kernels.



Source: docs/figures/14_kernel_round2_waterfall.png

Figure 8. Kernel-family roll-up before/after round 2

Read: Polar 73.9 -> 20.9 ms, backbone 19.4 -> 6.4 ms; tensor products stay at 10.8 ms.

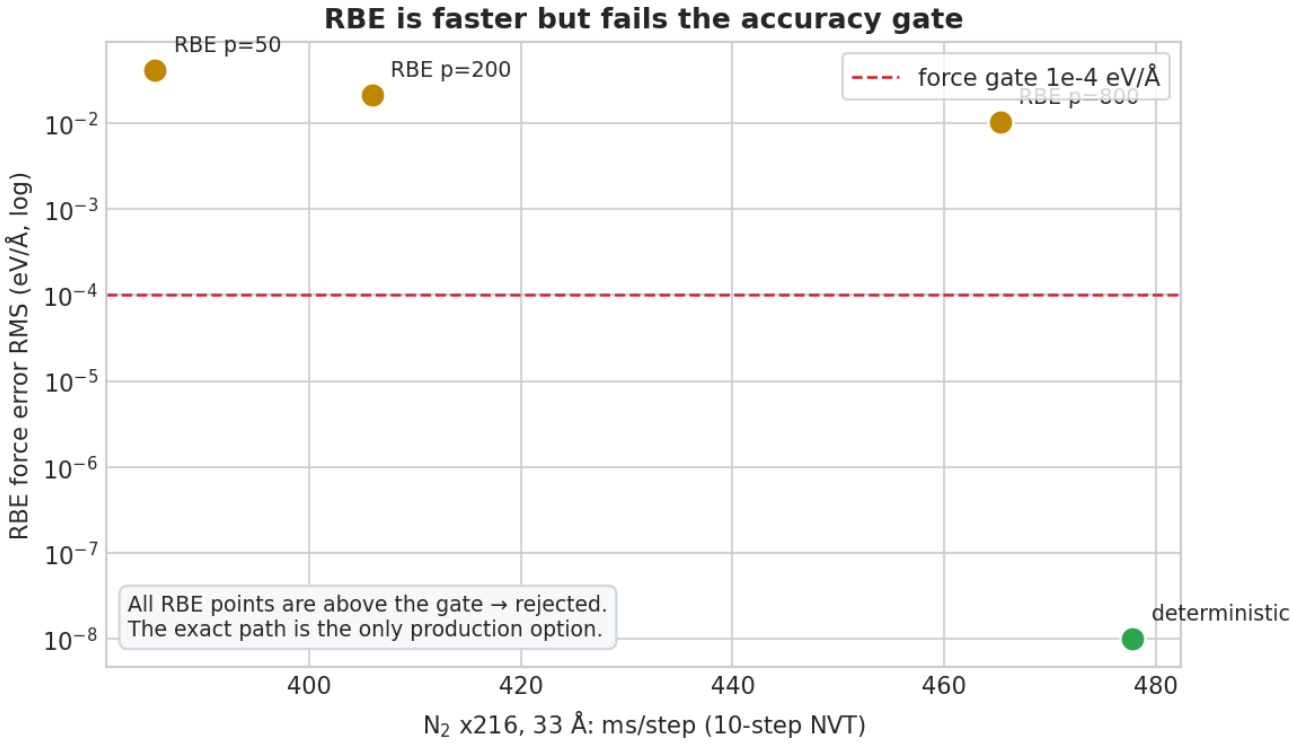


Source: docs/figures/15_kernel_round2_families.png

Accuracy, gates, and dynamics

Figure 9. RBE speed-accuracy tradeoff

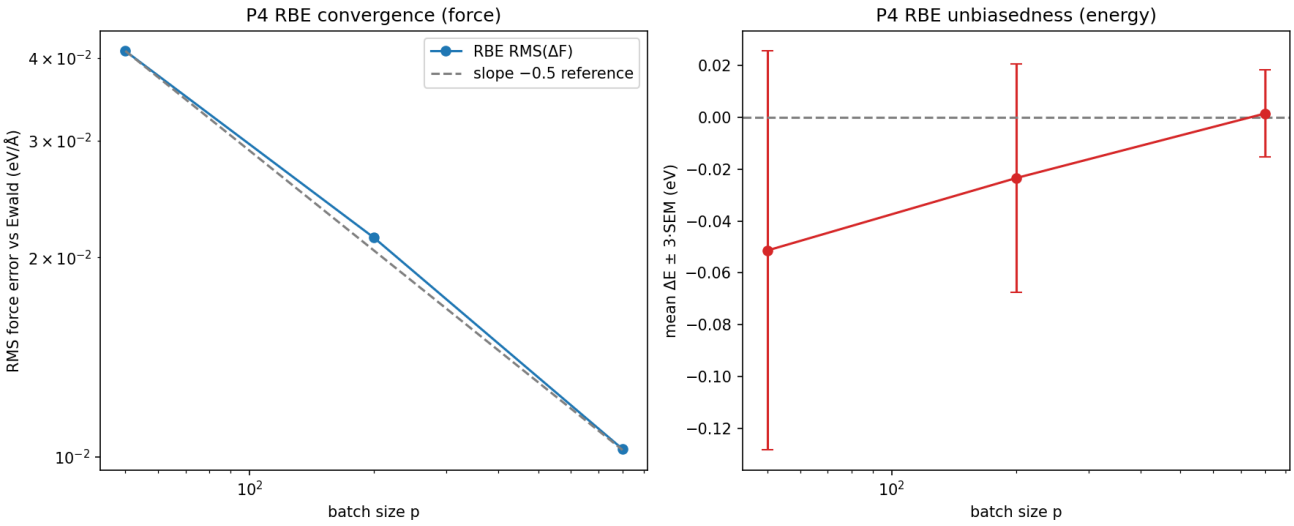
Read: RBE is faster but 100-400x over the 1e-4 eV/A force gate; rejected as production.



Source: docs/figures/02_rbe_tradeoff.png

Figure 10. RBE force error vs batch size

Read: Every displayed batch size is above the hard force gate.



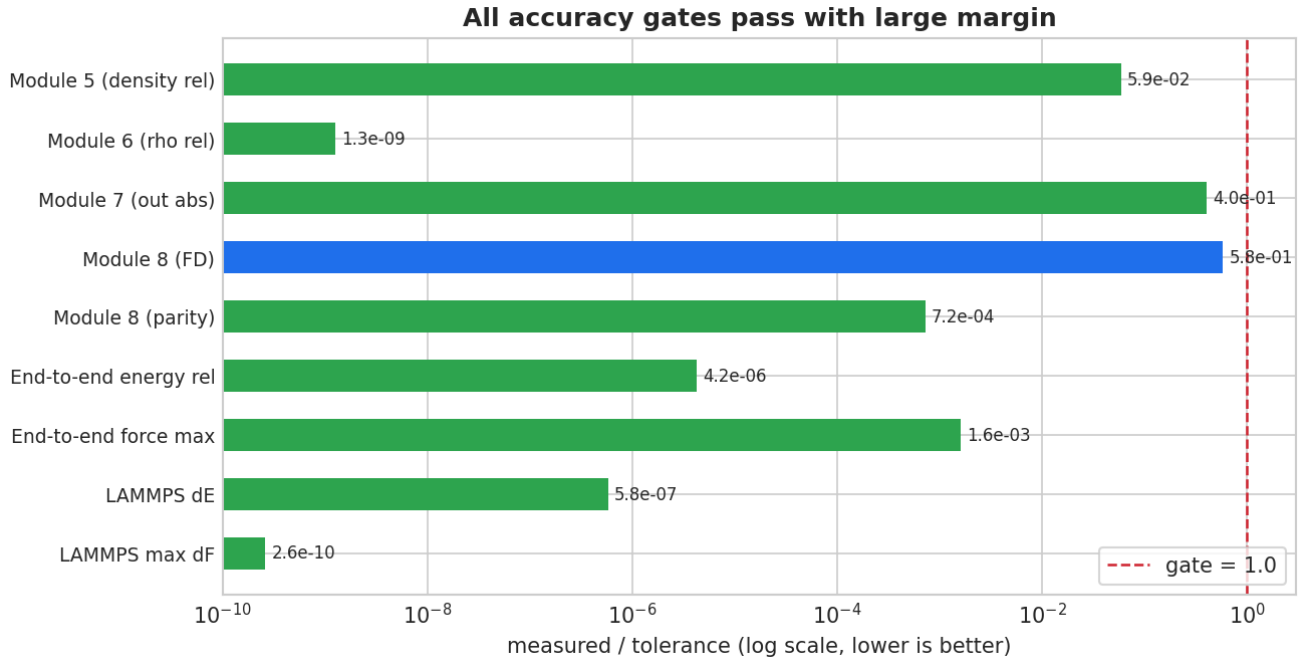
Source: results/polar_rbe_gate_v29.png

Figure 11. Accuracy gate margins

Read: All accepted gates pass with large margin; the plot is the measured/tolerance ratio.

Figure 11. Accuracy gate margins

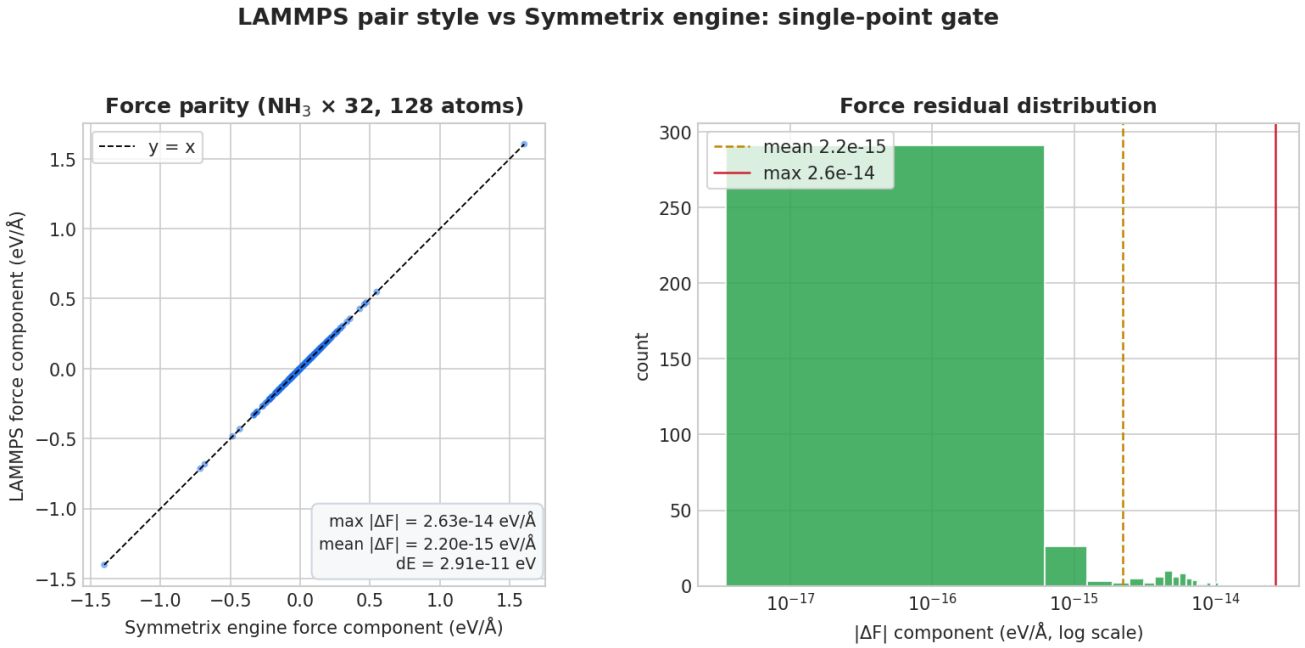
Read: All accepted gates pass with large margin; the plot is the measured/tolerance ratio.



Source: docs/figures/03_gate_margins.png

Figure 12. LAMMPS single-point force parity

Read: dE 2.9e-11 eV, max dF 2.6e-14 eV/A; the residual distribution is machine noise.



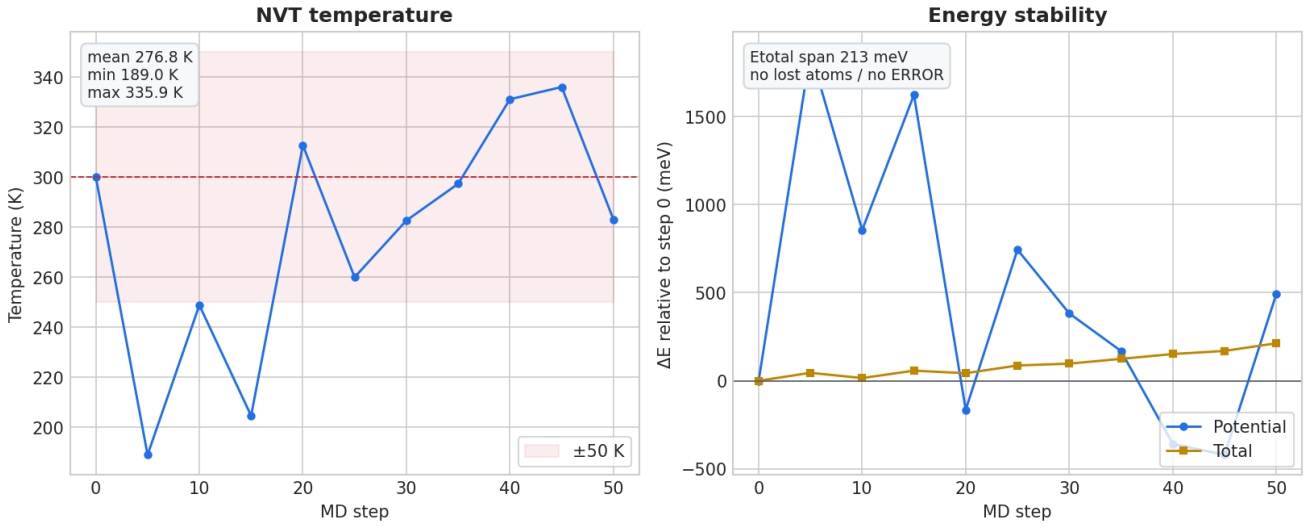
Source: docs/figures/07_lammps_parity.png

Figure 13. NVT MD stability (50 steps)

Read: No lost atoms; Etot span 0.21 eV; temperature/pressure remain stable.

Figure 13. NVT MD stability (50 steps)

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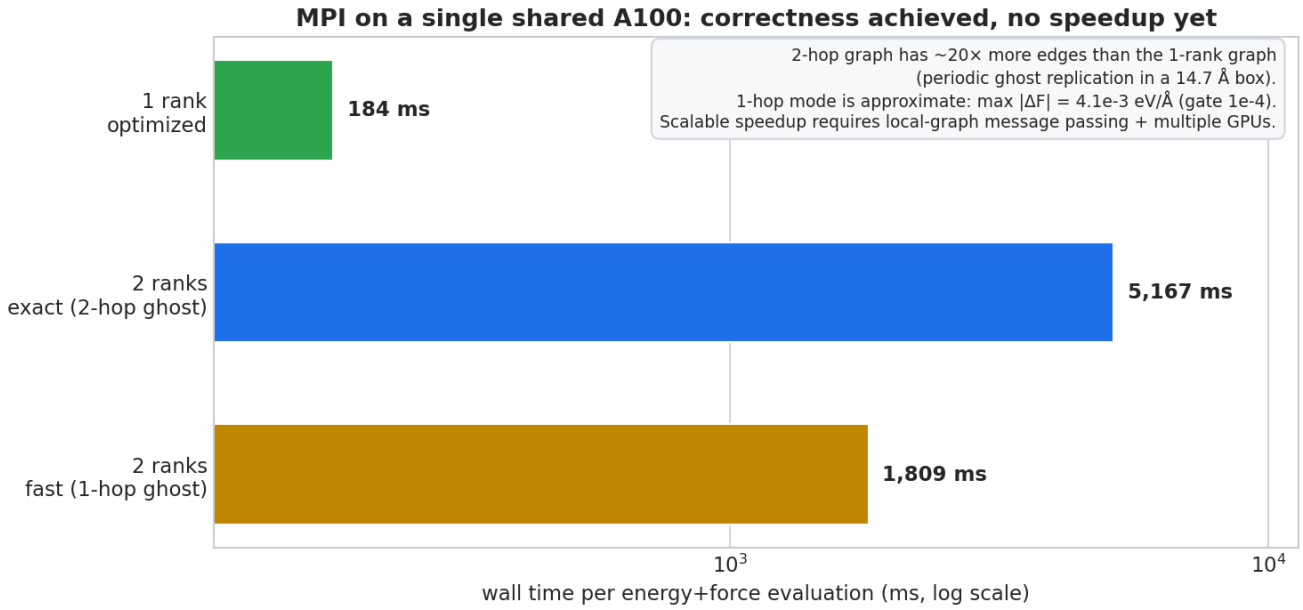


Source: docs/figures/08_nvt_md.png

MPI

Figure 14. MPI status: correctness achieved, no one-GPU speedup

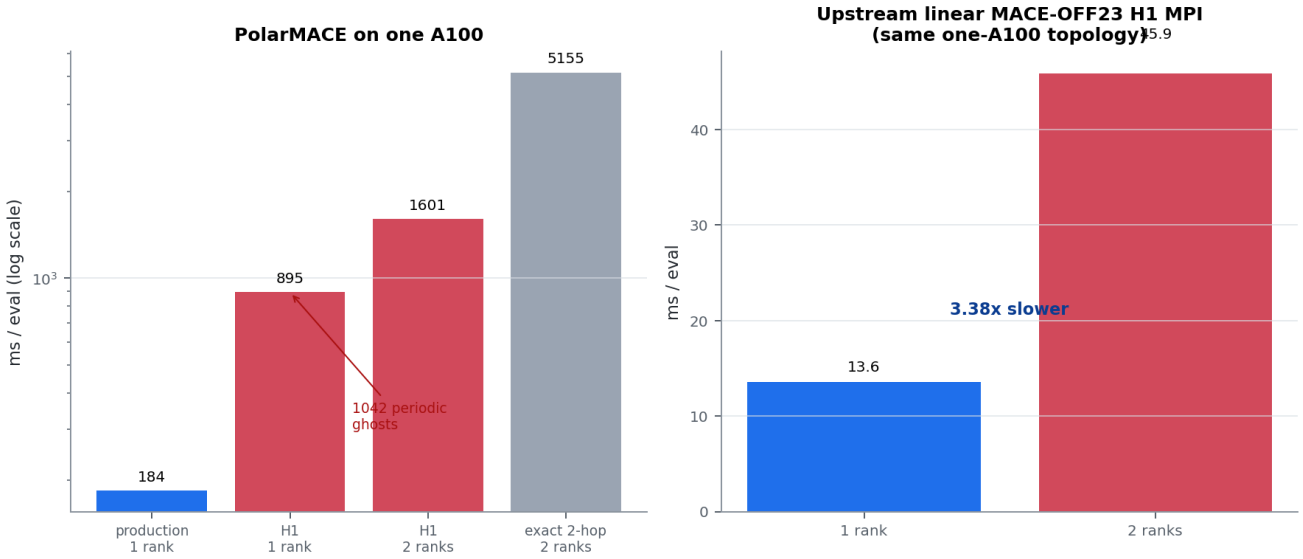
Read: Exact 2-hop and H1 pass the force gate; both are slower on one shared A100.



Source: docs/figures/06_mpi_status.png

Figure 15. MPI H1 pre-fix diagnosis: the 1042-ghost problem

Read: Historical: H1 1-rank ran 1170 nodes vs 128 (4.87x). The one-rank fold removed it: 418.64 -> 50.20 ms.



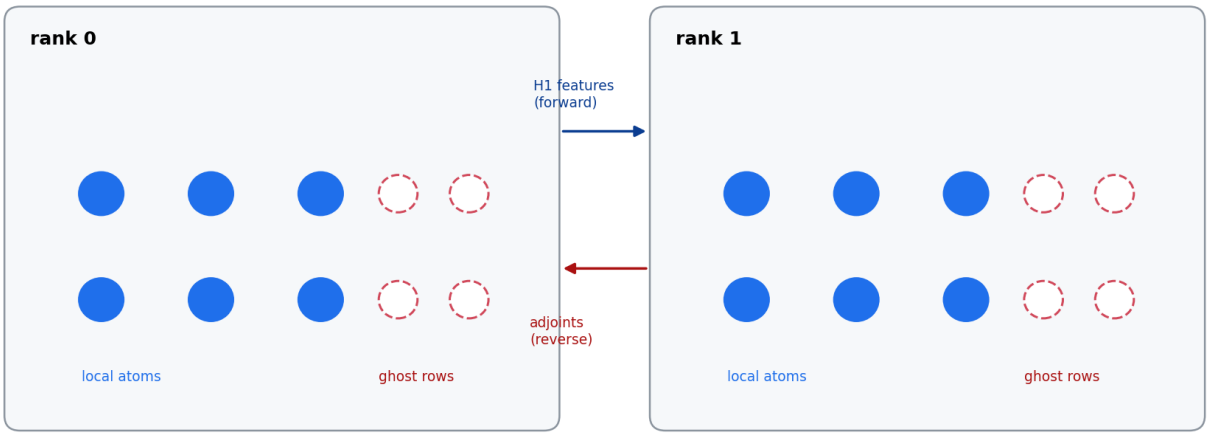
Source: docs/figures/17_mpi_h1_diagnosis.png

Figure 16. H1 forward/adjoint exchange schematic

Read: Forward fills ghost rows with H1; reverse sums ghost adjoints back to the owner.

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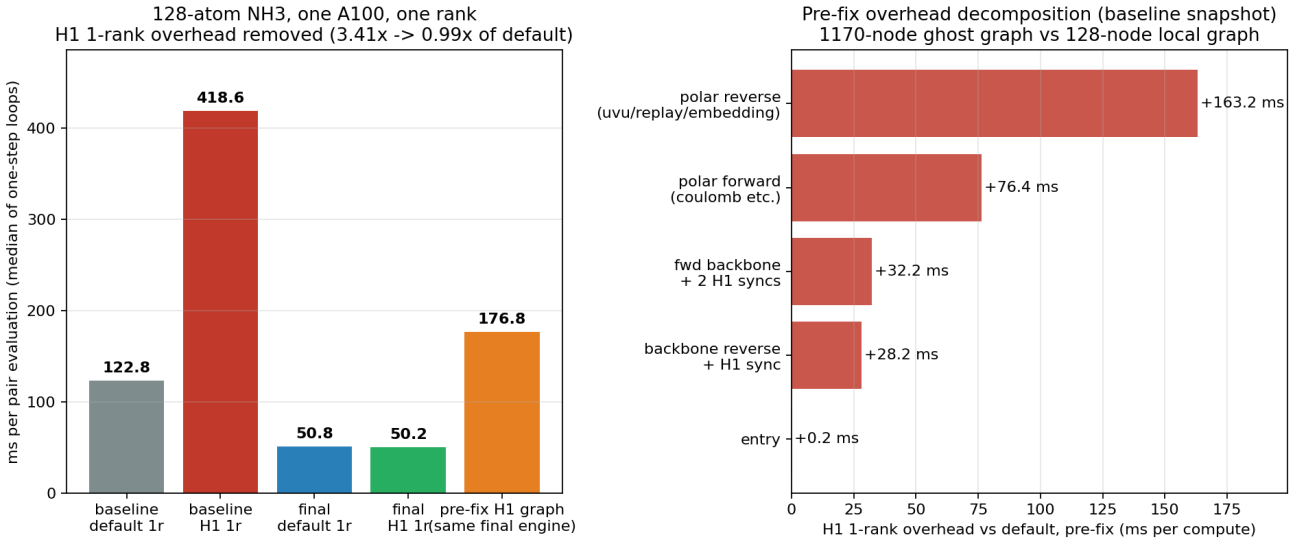


Each layer: exchange H1 (N x C) once; ghosts only need H1, not the full readout/polar pipeline.

Source: docs/figures/18_mpi_h1_schematic.png

Figure 17. MPI H1 final: one-rank fix and two-rank blocker

Read: H1 1-rank 50.20 ms vs default 50.84 (0.99x); 2-rank is blocked by six 2.0-5.4 GiB/rank workspaces (peak 80,966 MiB).

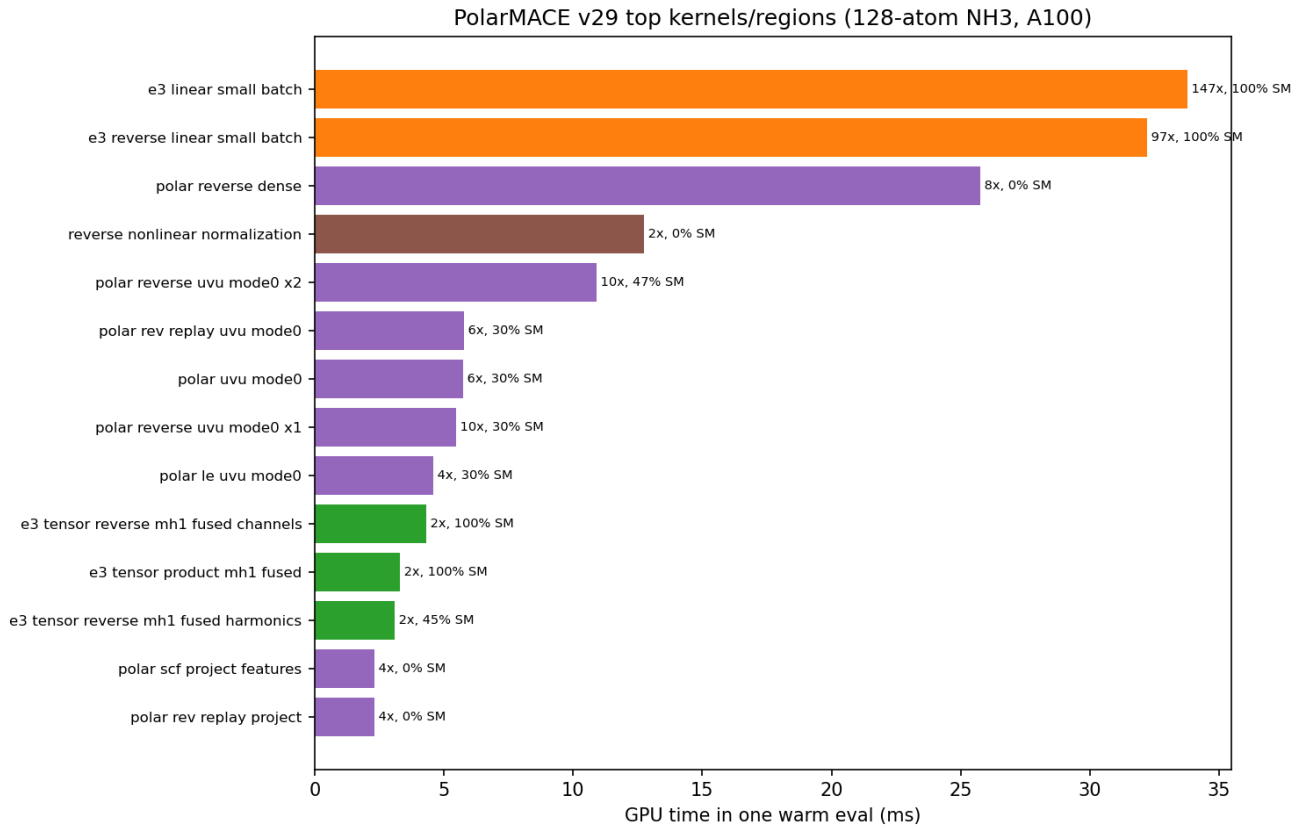


Source: docs/figures/19_mpi_h1_final.png

Kernel profile

Figure 18. Top GPU kernels by total time (pre-round-2 profile)

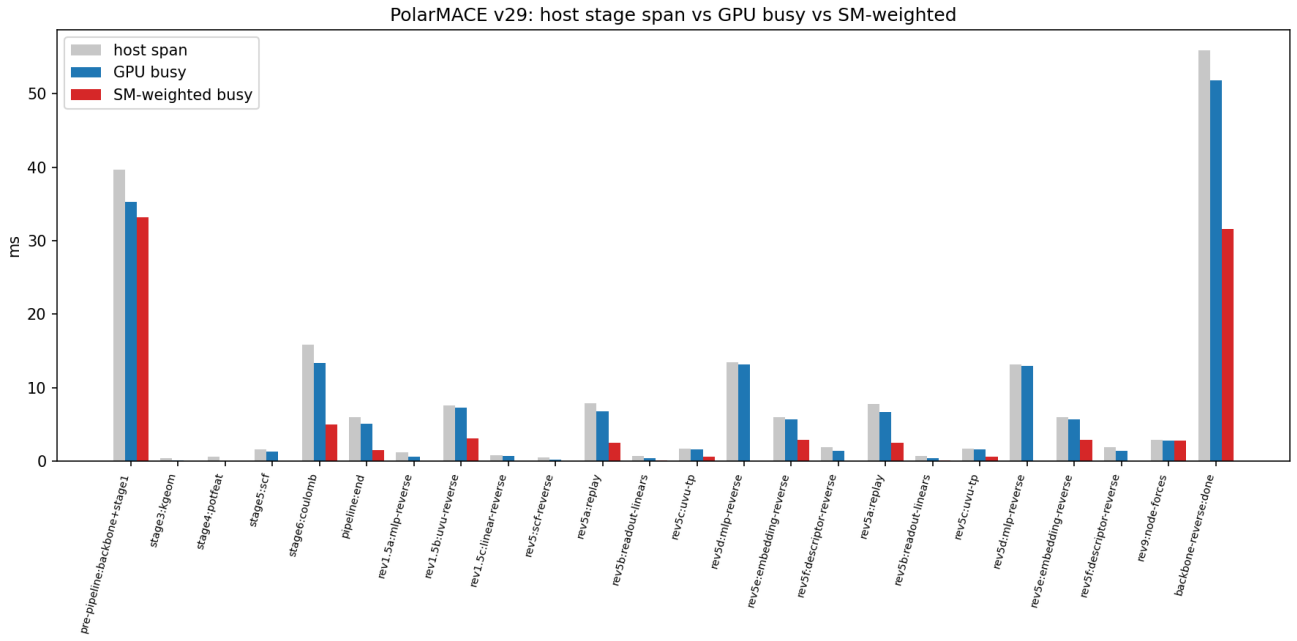
Read: The starting point: polar pipeline and e3 scalar linear dominated.



Source: docs/figures/09_kernel_top_labels.png

Figure 19. Host stage spans vs GPU busy

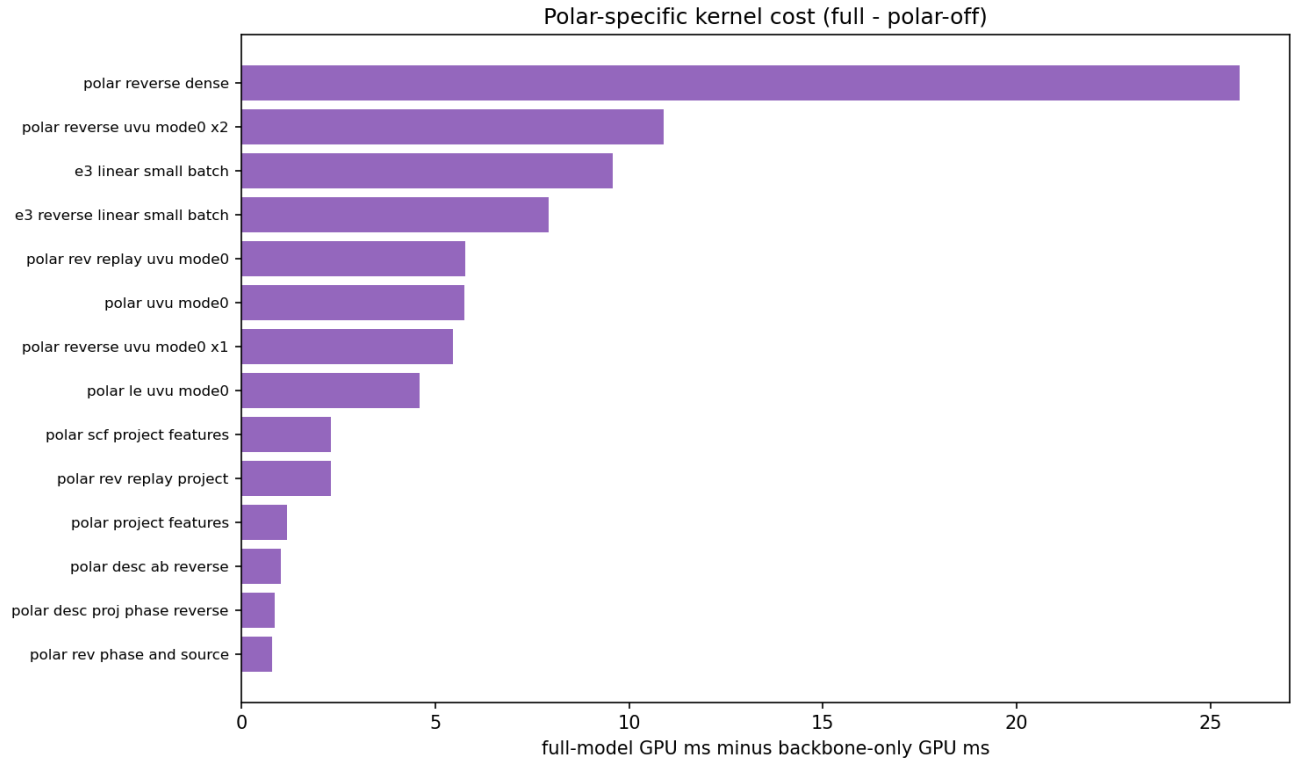
Read: How the eval's host stages map to measured GPU kernel time.



Source: docs/figures/10_kernel_stage_stack.png

Figure 20. Polar-specific kernel delta

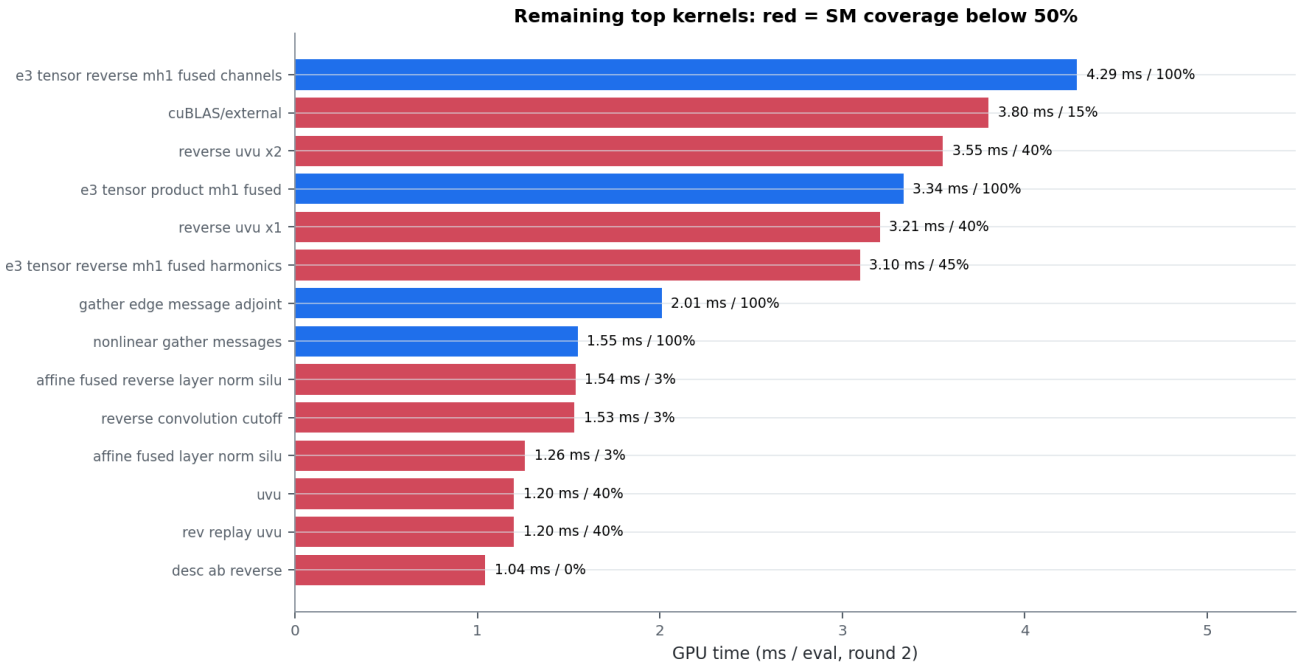
Read: The kernels the polar layer adds relative to the polar-off backbone.



Source: docs/figures/11_kernel_polar_delta.png

Figure 21. Remaining kernels after round 2 and their SM coverage

Read: Red = below 50% static coverage. Tensor products are 100% covered but instruction/store-bound; see the round-3 plan.



Source: docs/figures/16_kernel_round2_coverage.png

Theory formulas

The formulas behind the energy model, tensor algebra, performance model, and MPI design. Full derivations and the per-gate test catalog are in docs/THEORY_AND_METHODS.md.

Equation 1. GTO-Ewald reciprocal-space energy

Read: The Gaussian is inside the GTO form factor; do not apply it twice.

$$E_{\mathrm{lr}} = \frac{2\pi}{V} \sum_{\mathbf{k} \neq 0} \frac{|\tilde{\rho}(\mathbf{k})|^2}{k^2}, \qquad \tilde{\rho}(\mathbf{k}) = \sum_j \tilde{\rho}_j(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}_j}$$

Source: docs/figures/eq_ewald.png

Equation 2. Self-consistent charge/spin density fixed point

Read: The engine iterates the Fukui-balanced update to a density tolerance.

$$\rho^{(t+1)} = (1-\beta)\rho^{(t)} + \beta\,\mathcal{F}(\rho^{(t)}), \qquad \|\rho^{(t+1)} - \rho^{(t)}\| < \varepsilon$$

Source: docs/figures/eq_scf.png

Equation 3. e3nn linear layer

Read: A batched matrix multiply; round 1 routed it through packed FP64 GEMM.

$$y_j = \sum_i W_{ji} x_i + b_j$$

Source: docs/figures/eq_linear.png

Equation 4. e3nn tensor product

Read: The Clebsch-Gordan contraction implemented by the tensor-product kernels.

$$z_m^{(l)} = \sum_{l_1 m_1 l_2 m_2} C^{lm}_{l_1 m_1 l_2 m_2}\, x_{m_1}^{(l_1)}\, y_{m_2}^{(l_2)}$$

Source: docs/figures/eq_tensor.png

Equation 5. Static SM coverage and SM-weighted busy time

Read: The definition behind every occupancy number in this report; it is a static estimate.

$$O_i = \min\left(1, \frac{G_i}{N_{\mathrm{SM}} B_i}\right), \quad T_{\mathrm{SM}} = \sum_i t_i O_i$$

Source: docs/figures/eq_occupancy.png

Equation 6. Amdahl speedup and ceiling

Read: Explains why a 12x kernel gain becomes a 1.5x end-to-end gain.

$$S_{\mathrm{tot}} = \frac{1}{(1-p) + p/s}, \quad S_{\max} = \frac{1}{1-p}$$

Source: docs/figures/eq_amdahl.png

Equation 7. H1 communication volume and multi-rank wall-time model

Read: Bytes per exchange are small; the 1-rank overhead is ghost compute, not communication.

$$V_{\mathrm{comm}} = N_g C b, \quad T_r \approx \frac{T_{\mathrm{local}} + T_{\mathrm{comm}}}{r} + T_{\mathrm{global}}$$

Source: docs/figures/eq_comm.png